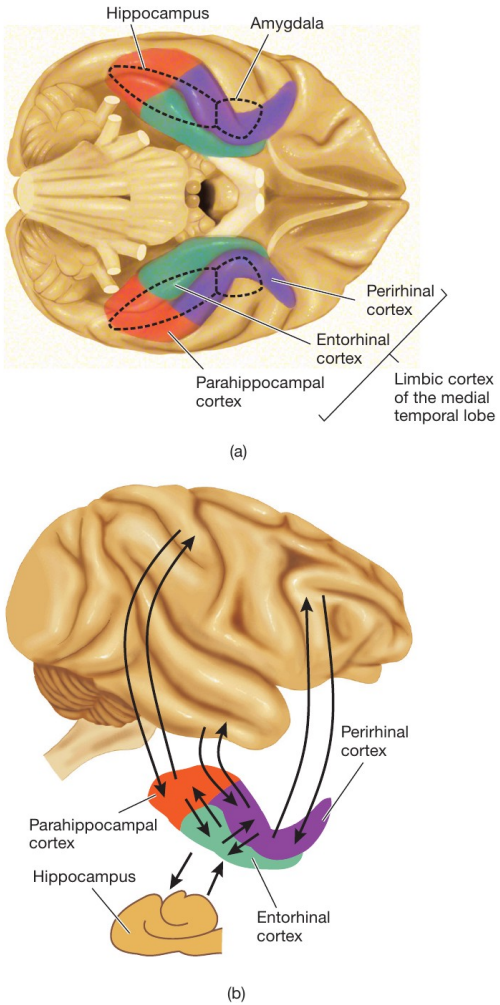


Carlson and Birkett (2022), Fig. 13.20, p. 434.

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Learning Systems, Memory Circuits, and Lasting Synaptic Change

From Experience to Memory: Learning, Plasticity, and Recall

Fall 2026

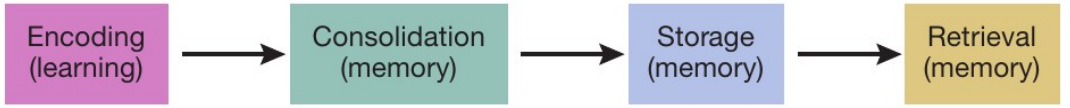
Jordan Lopez, Ph.D.
Professor of Neurobiology & Biomedical Engineering

Learning changes what later experience can produce

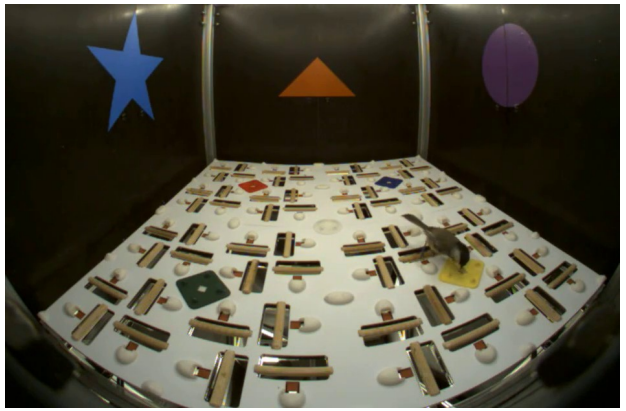
Learning is a change produced by experience in how an organism responds or processes information. Memory is the persistence of information that can influence a later response. These processes involve changes in neural activity and cellular properties. A measured improvement in behavior is evidence of learning, while identifying its biological storage requires additional experiments.

Encoding refers to the initial processing that establishes information about an experience. Consolidation refers to processes that help stabilize or reorganize that information over time. Retrieval is the use of retained information when an appropriate cue or situation occurs. A failure at any one of these stages can reduce later performance even if the other stages remain functional.

Behavior also depends on sensation, motivation, attention, and movement. An animal that fails to approach a remembered reward might be unable to detect the cue or no longer value the outcome. Memory experiments therefore compare responses under conditions that help separate retained information from the other requirements needed to express it.



Carlson and Birkett (2022), Fig. 13.2, p. 417.



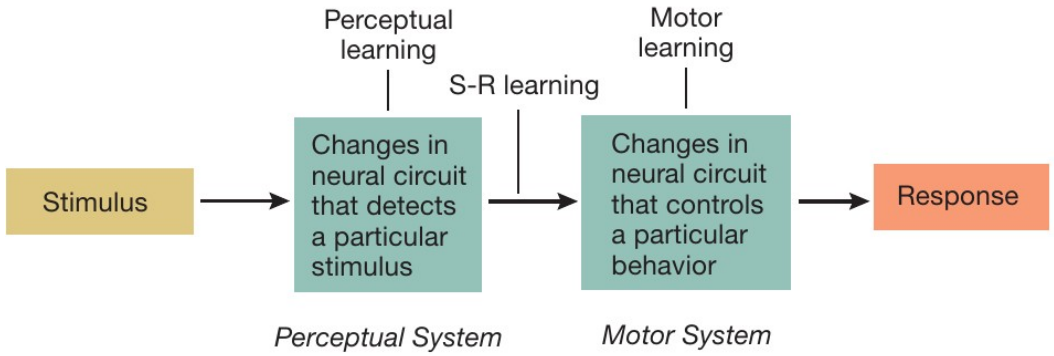
Applegate et al. (2022), Video 1.

Different learning tasks change different neural operations

Perceptual learning changes the ability to recognize or discriminate stimuli through experience. Stimulus–response learning changes the action evoked by a cue or the consequences associated with an action. Motor learning improves the organization of movement. Relational learning represents relationships among events, places, or other pieces of information rather than only one isolated cue and response.

These categories can overlap within a single task. Learning to find a food location may require recognizing landmarks, selecting a route, and improving movement through the environment. The experimenter can alter task demands to emphasize one component. Performance on one task therefore cannot be treated as a complete measurement of every form of learning.

Different neural systems make different contributions to these operations. Sensory cortex can change stimulus representations, corticostriatal circuits can alter action selection, and hippocampal networks can represent relationships and context. These systems interact, but their contributions can be dissociated when damage or selective manipulation impairs one kind of learning while leaving another relatively preserved.



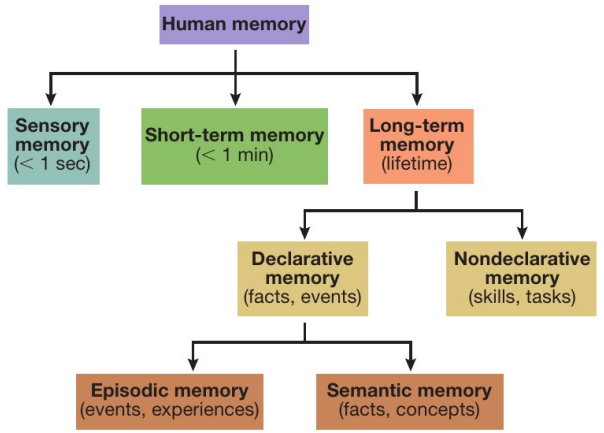
Carlson and Birkett (2022), Fig. 13.6, p. 420.

Conscious recollection can fail while skill learning survives

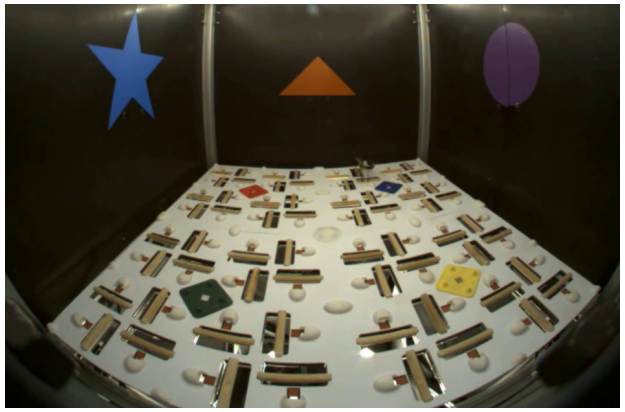
Declarative memory includes information that can be consciously recalled, such as an event or a fact. Nondeclarative memory includes skills, habits, and other effects of experience that need not be consciously described. The distinction concerns how information is expressed and used, rather than whether learning caused a physical change somewhere in the nervous system.

Henry Molaison, known in the scientific literature as H. M., developed severe difficulty forming new declarative memories after extensive bilateral medial-temporal surgery. The affected tissue extended beyond a small isolated hippocampal lesion. He could nevertheless improve on some motor and perceptual tasks without remembering the practice episodes that produced the improvement.

His immediate conversation and some short-term retention were also more preserved than his ability to form lasting new event memories. The pattern showed that retaining a small amount of current information and establishing a later recollection are different operations. A memory disorder can therefore be profound without eliminating all learning, intelligence, or moment-to-moment awareness.



Carlson and Birkett (2022), Fig. 13.9, p. 422.



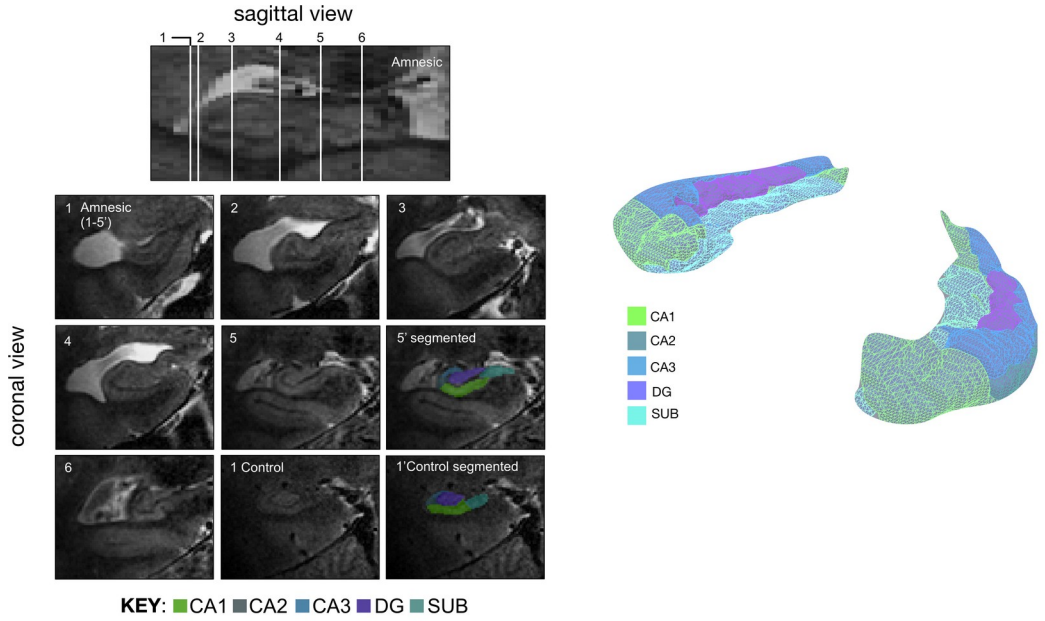
Applegate et al. (2022), Video 2.

Human imaging localizes hippocampal subfield damage

A human study used high-resolution magnetic resonance imaging to measure hippocampal subfields after limbic encephalitis. Subfields are anatomically distinguishable components, including CA1, CA3, and the dentate gyrus. The investigators traced these regions along the hippocampus to identify which portions showed persistent volume loss after the illness.

The group showed prominent bilateral CA3 damage. Quantitative segmentation provided a more specific description than simply labeling the entire medial temporal lobe as abnormal. This localization made it possible to compare the memory profile with a particular pattern of surviving and damaged tissue, rather than attributing every symptom to an unspecified global loss of brain function.

Imaging measurements identify anatomy, while interviews and behavioral tasks identify what a person can remember. Their combination allows researchers to ask whether a relatively localized structural difference is associated with a specific change in recollection. The anatomical measurement alone does not reveal the content of a stored memory or the activity of every surviving neuron.



7.0-Tesla 3D fast-spin echo MR images (basis for whole-hippocampus subfield quantitative morphometry)

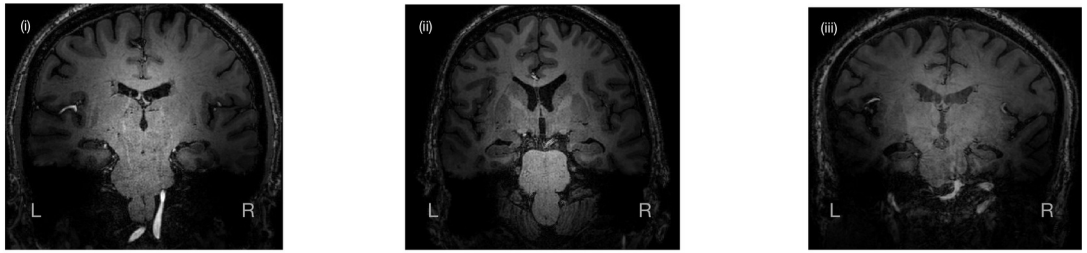
Miller et al. (2020), Fig. 1.

Localized CA3 loss changes selected memory functions

The CA3 study also examined whole-brain anatomical images to evaluate the extent of volume loss. This analysis assessed whether the hippocampal abnormality was part of an obvious widespread pattern. The observed localization helped constrain the explanation of the memory findings, although anatomical imaging and microscopic cellular integrity remain different levels of measurement.

Participants underwent detailed memory assessment rather than being classified as having no memory at all. The investigators distinguished episodic details about a particular event from more general information and other cognitive abilities. Such comparisons identify what is selectively difficult while documenting capacities that remain available after the illness.

This approach differs from assuming that one brain structure is required equally for every memory task. A patient may retain general knowledge, some immediate information, and familiar skills while losing details needed to mentally revisit an event. The biological account must explain that pattern of preservation and impairment, not merely the presence of a lesion.



7.0-Tesla T_1-weighted anatomical images used as the basis for whole-brain voxel-based morphometry

Miller et al. (2020), Fig. 3.



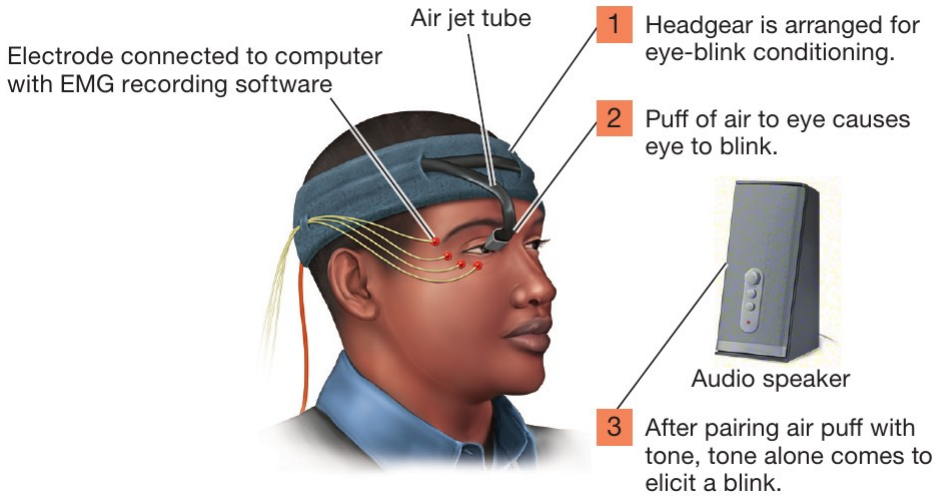
Krishnan et al. (2025), Video 1.

Conditioning changes what a predictive cue evokes

Classical conditioning occurs when a stimulus acquires the ability to evoke a response through its relationship with another event. In eyeblink conditioning, an air puff naturally produces a blink. A tone presented before the puff can become a conditioned stimulus, eventually producing a blink before the puff arrives.

The air puff is called the unconditioned stimulus because it can produce the response without that pairing history. The learned tone-evoked blink is a conditioned response. Learning changes how sensory information influences an existing response system; it does not require the animal to invent a completely new muscle or sensory receptor.

The timing and predictive relationship between events influence what is learned. A tone that consistently precedes the puff differs from a tone that occurs randomly relative to it. Experimental comparisons manipulate these relationships to determine whether behavior reflects an acquired association rather than repeated exposure, generalized arousal, or adaptation to the unconditioned event.



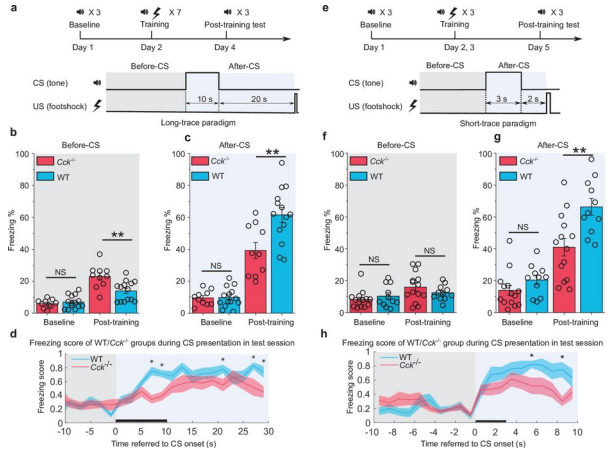
Carlson and Birkett (2022), Fig. 13.3, p. 418.

A temporal gap reveals a role for peptide signaling

A mouse study tested trace fear conditioning, in which a tone ends before an aversive event begins. The interval requires information about the earlier cue to remain relevant across time. The investigators compared normal mice with animals lacking cholecystokinin, or CCK, a peptide that can modulate neural communication in several brain regions.

Mice lacking CCK showed reduced later conditioned freezing in the trace arrangement. The study compared learning conditions and measured the response to the trained cue. The result identified a contribution of peptide signaling to the formation of this temporally separated association, rather than showing that every movement or sensory response required CCK.

The preparation also allowed the researchers to examine a pathway from entorhinal cortex toward the lateral amygdala. This route could provide information relevant to the interval between cue and outcome. Anatomical specificity and task timing helped identify the conditions under which the peptide contribution affected the learned defensive response.



Feng et al. (2021), Fig. 1.



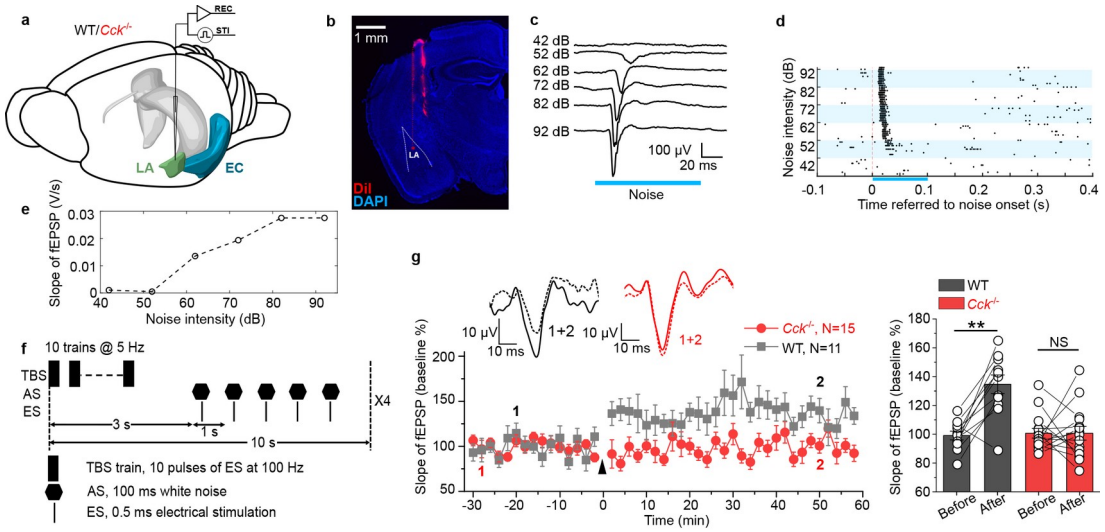
Feng et al. (2021), Video 1.

Replacing a missing peptide can restore local plasticity

The trace-conditioning study recorded responses in the lateral amygdala while testing long-term changes in neural communication. High-frequency stimulation normally produced a lasting increase in the measured response. This potentiation was impaired in mice lacking CCK, providing a cellular comparison alongside the behavioral learning deficit.

Applying CCK restored the relevant plasticity response under the experimental conditions. The rescue showed that the tissue retained machinery capable of expressing the change when the missing modulatory signal was supplied. A lifelong genetic deletion had therefore not simply destroyed every synapse involved in the recorded pathway.

The investigators combined peptide replacement with pathway manipulations and learning measurements. These experiments distinguished a permissive signaling contribution from the sensory identity of the tone or aversive event. CCK did not itself specify the entire remembered experience; it altered conditions under which activity could produce a lasting change in the tested circuit.



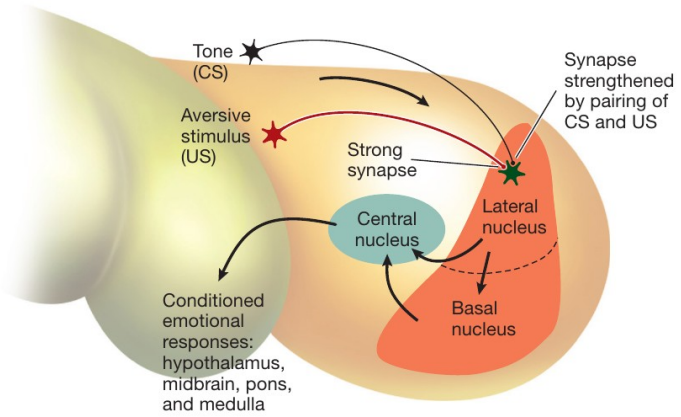
Feng et al. (2021), Fig. 2.

Associations change sensory access to response circuits

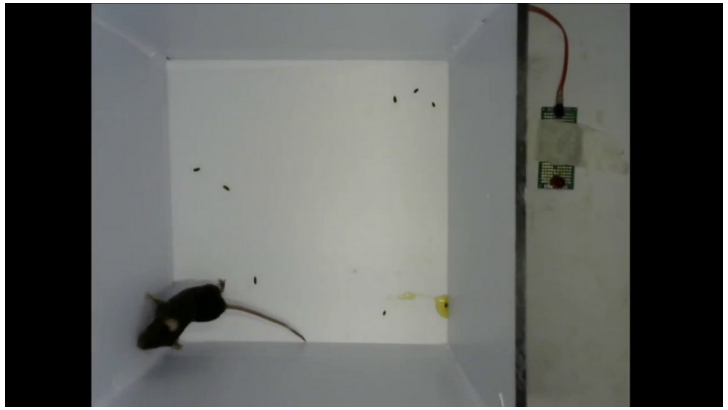
In conditioned defensive learning, sensory inputs about a cue and an aversive event converge within amygdala-related networks. Plasticity can increase the effectiveness of the cue pathway, allowing it to recruit downstream responses later. The original event and the later predictive cue need not activate identical sensory receptors to influence a shared response system.

Glutamate receptors contribute to changes at many of these synapses. AMPA receptors carry fast excitatory current, while NMDA receptors can permit calcium entry when transmitter binding coincides with sufficient postsynaptic depolarization. This coincidence-sensitive property helps explain how the timing of converging activity can influence which synapses become stronger.

The learned response also depends on context and the animal's current state. A tone learned in one environment may evoke a different response in another. Experiments that manipulate the context can therefore distinguish a simple cue association from a representation that includes where the event occurred and which surrounding conditions predicted its consequences.



Carlson and Birkett (2022), Fig. 13.11, p. 424.



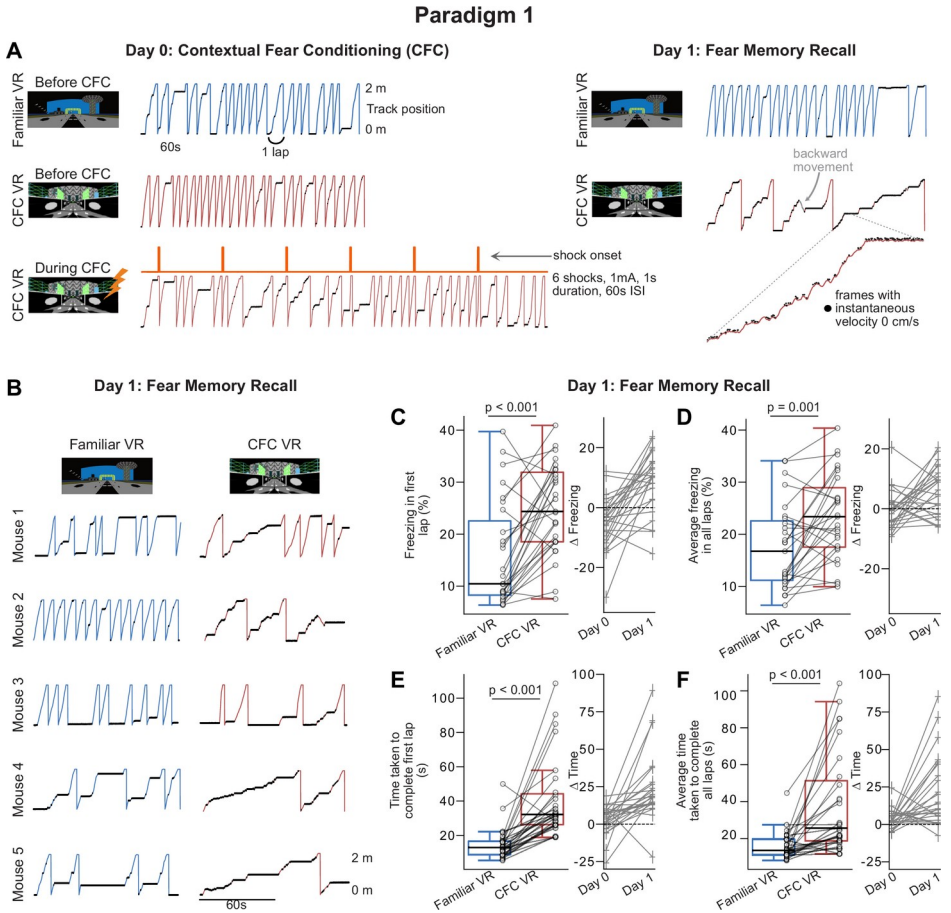
Feng et al. (2021), Video 2.

Virtual contexts support defensive learning

A recent mouse study developed contextual fear conditioning during head-fixed virtual navigation. Mice moved on a treadmill while surrounding displays changed with their movement. An aversive event was paired with one virtual environment, allowing later behavior in that environment to be compared with behavior in a neutral setting.

After conditioning, mice showed increased freezing-like immobility and altered running in the shock-paired environment. The strongest differences depended on the tested phase and paradigm. These behavioral changes demonstrated that a head-fixed animal could acquire an environment-specific association while the apparatus also permitted stable imaging of hippocampal neurons.

Freezing was defined from movement on the treadmill rather than from unrestricted whole-body behavior in an open chamber. The operational definition matters because the animal's head was fixed. The preparation made a particular form of contextual learning accessible to high-resolution recording, while preserving a clear description of what the measured response actually was.



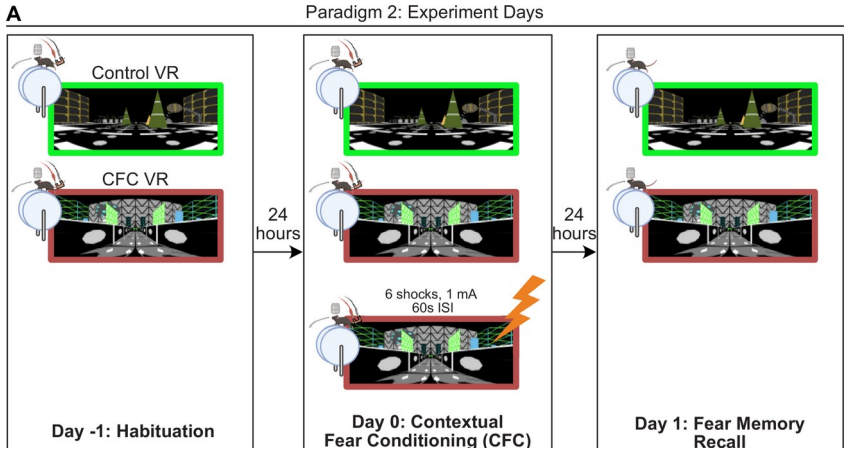
Krishnan et al. (2025), Fig. 2.

A different neutral context changes the comparison

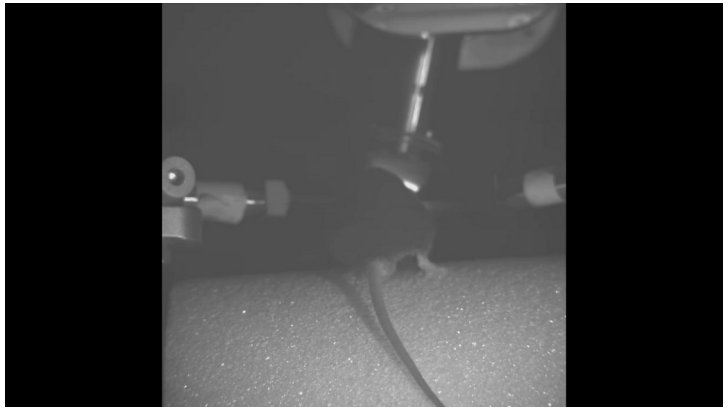
The virtual-conditioning study compared several versions of the task. In one version, the neutral environment was already familiar, whereas another used a novel neutral environment. The researchers varied these conditions to determine how the comparison context influenced later differences in freezing and running after aversive learning.

A novel neutral environment introduced its own changes in exploration and movement. The magnitude of the difference between the aversive and neutral contexts therefore depended on more than the shock pairing alone. The experiment measured behavior before and after conditioning in both settings, allowing familiarity and learned threat to be evaluated separately.

Later refinements also altered the sensory conditions carried into the recall session. These changes affected how consistently the memory was expressed. The study demonstrates a concrete retrieval principle: the cues and bodily conditions present during the test can change measured recall, even when the earlier learning procedure included the same type of aversive outcome.



Krishnan et al. (2025), Fig. 3.



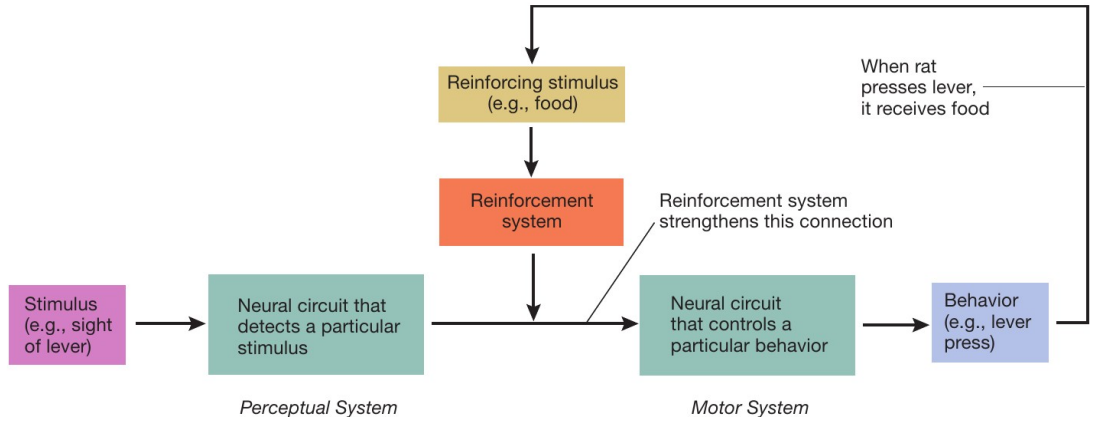
Krishnan et al. (2025), Video 2.

Operant learning changes actions through consequences

Operant conditioning occurs when the consequences of an action alter its later probability. A rat may learn to press a lever because pressing delivers food. The response is selected from behavior the animal can perform, and repeated consequences change how often that action is chosen under the relevant conditions.

Positive reinforcement adds an outcome that increases behavior, while negative reinforcement removes or prevents an aversive event and also increases behavior. Punishment decreases behavior. The word negative therefore refers to removal, not to whether learning is unpleasant or harmful. Distinguishing these terms identifies what changed in the consequence and which direction the action changed.

Timing, contingency, and the value of the outcome influence learning. An action reliably followed by food differs from an action followed by food regardless of whether it occurred. Tests that change the outcome's value or the action–outcome relationship help determine whether behavior still uses its consequence or has become relatively habitual.



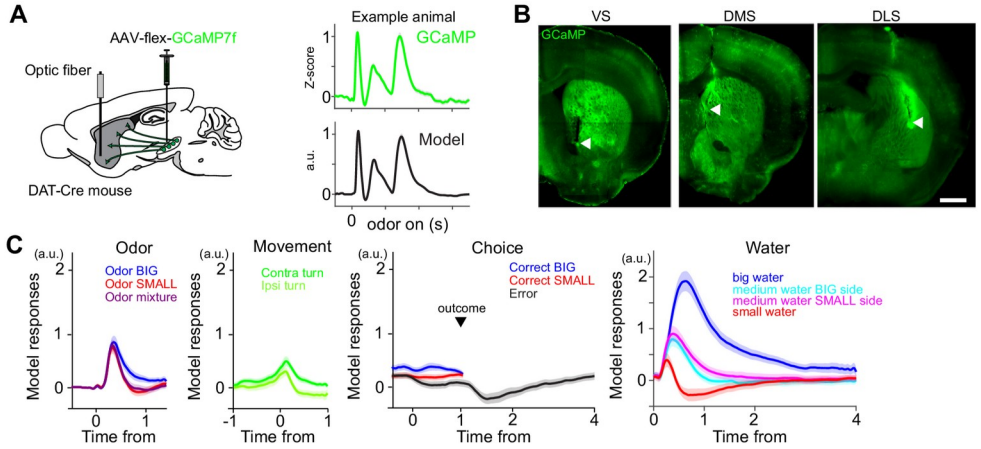
Carlson and Birkett (2022), Fig. 13.5, p. 419.

Dopamine axons respond to changes in expected reward

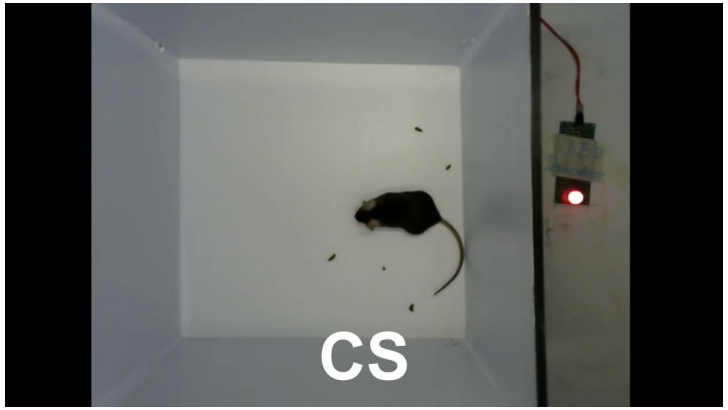
A mouse study recorded calcium activity from dopamine axons in ventral, dorsomedial, and dorsolateral striatum during a decision task. Different trial events provided information about possible reward. The researchers asked whether the axonal signals reflected only reward delivery or also changes in what the animal could predict before the outcome occurred.

The signals varied around cues, choices, and outcomes in ways consistent with changes in expected reward. A temporal-difference model described the error between successive reward expectations as the task unfolded. This interpretation differs from labeling every rise in dopamine-related activity as an immediate experience of pleasure.

The experiment measured calcium in identified axons rather than directly counting every released dopamine molecule. Comparing task events and model predictions helped identify the information represented by the signal. The relevant teaching point is the sequence: a cue changes expectation, an outcome confirms or violates it, and the resulting signal can influence later learning.



Tsutsui-Kimura et al. (2020), Fig. 2.



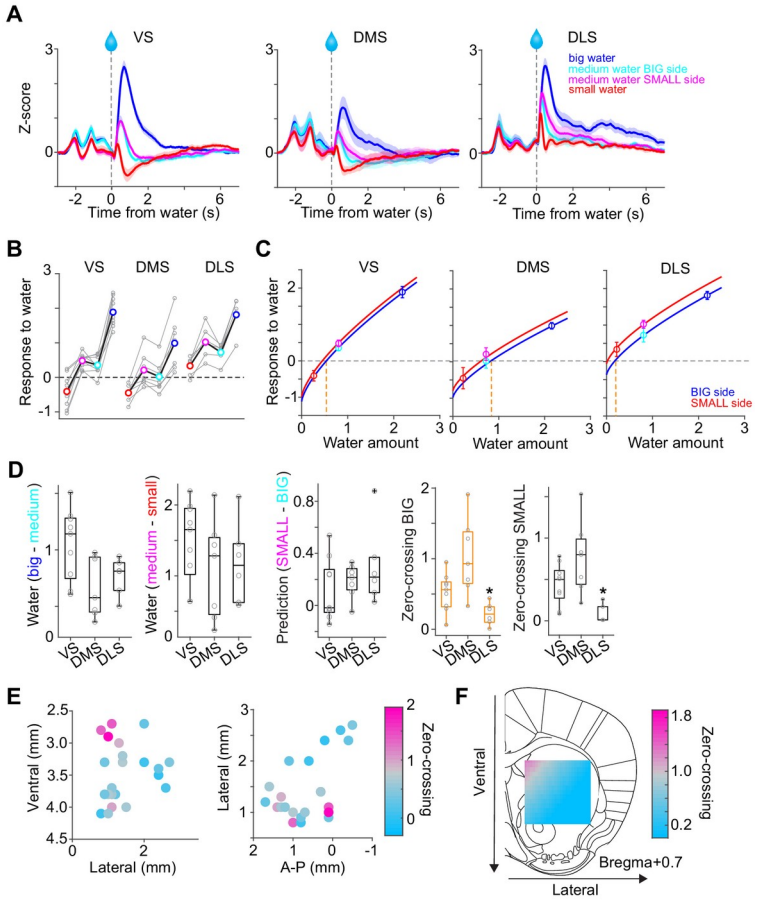
Feng et al. (2021), Video 3.

Reward responses differ across striatal territories

The dopamine-axon study compared responses to water outcomes across striatal regions. A reward can be better or worse than expected depending on the preceding cue and trial history. The same physical drop of water can therefore have a different learning significance under different expectations.

Many response features were similar across the three recorded territories, but dorsolateral reward responses were shifted upward. In that region, the signals lacked the same inhibitory response to negative prediction errors observed elsewhere. The difference placed a constraint on how one computational description could be implemented across anatomically distinct dopamine projections.

A negative prediction error is not the absence of all neural activity. It refers to an outcome or updated expectation that is worse than predicted. Identifying its neural representation requires a comparison with expectation and baseline. Region-specific measurements prevent the assumption that every dopamine projection carries the same range and sign of teaching signal.



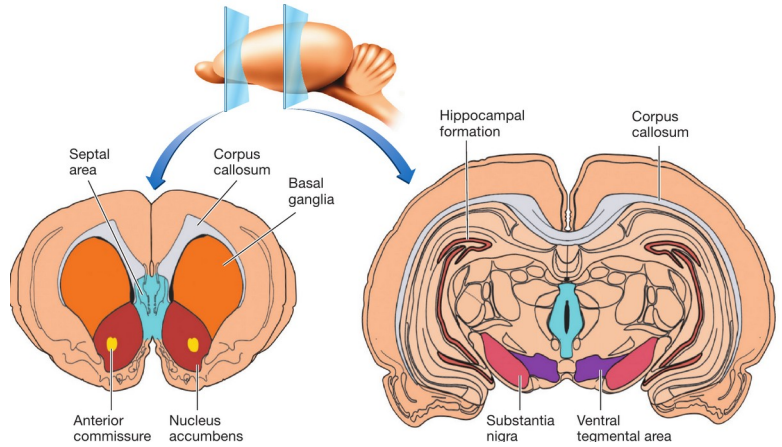
Tsutsui-Kimura et al. (2020), Fig. 4.

Reinforcement modifies pathways used for an action

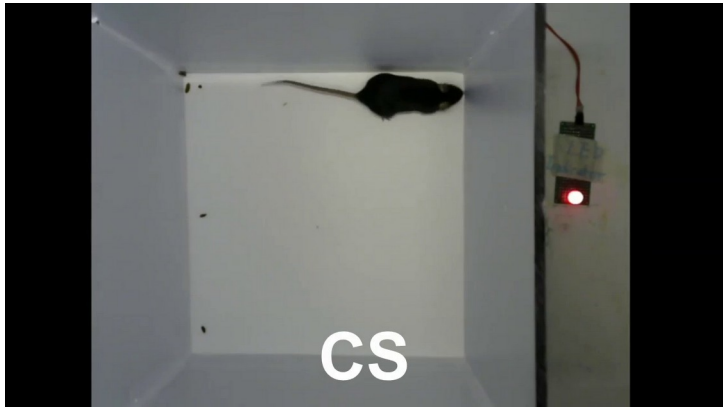
Reinforcement-related signals influence synapses within circuits that select and execute actions. Cortical inputs convey sensory, contextual, and movement-related information to the striatum. Dopamine can change how activity in these pathways produces plasticity, allowing recently active connections to be modified according to the consequence that follows.

The basal ganglia contain multiple interacting loops rather than one channel dedicated to reward. Different territories contribute to action selection, motor organization, and learned strategies. The effect of dopamine depends on the receptor-bearing cells and the timing of the signal relative to activity in the relevant corticostriatal synapses.

Learning which action earned an outcome is a credit-assignment problem. If several choices and events preceded a reward, the nervous system must determine which should be updated. Current knowledge about the task can influence this assignment, so reinforcement learning need not depend only on blindly increasing the value of the last movement performed.



Carlson and Birkett (2022), Fig. 13.15, p. 427.



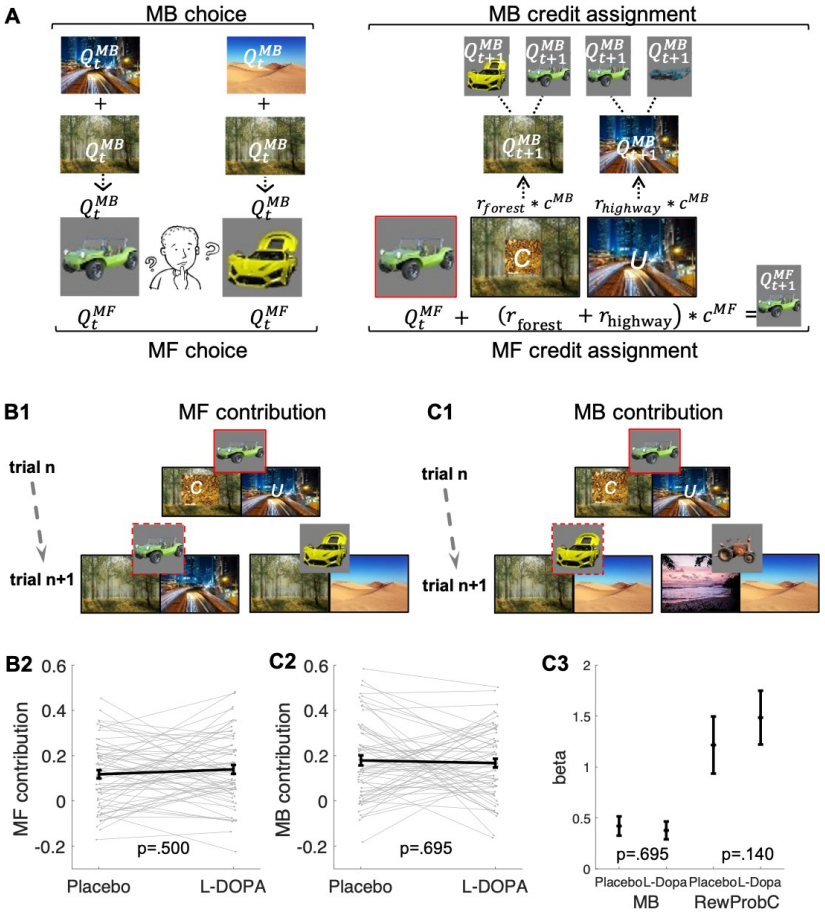
Feng et al. (2021), Video 4.

Human choices use learned values and task knowledge

A human study examined cooperation between model-free and model-based learning. Model-free values summarize previous reinforcement without explicitly calculating the full structure of the task. Model-based inference uses knowledge about how choices lead to later states and outcomes. Both processes can contribute to the value assigned to an option.

The task included situations in which the participant had to infer which earlier choice deserved credit for a later result. Behavior was compared with models that separated direct value-based choice from the use of task structure. This made it possible to test cooperation between the processes rather than forcing every decision into one exclusive category.

The investigators also used levodopa, a precursor that increases dopamine availability, in a controlled within-person comparison. This intervention tested how changing the dopamine-related condition affected the assignment of reward to prior choices. The pharmacological manipulation operated broadly, while the behavioral model identified a particular computational component that could change.



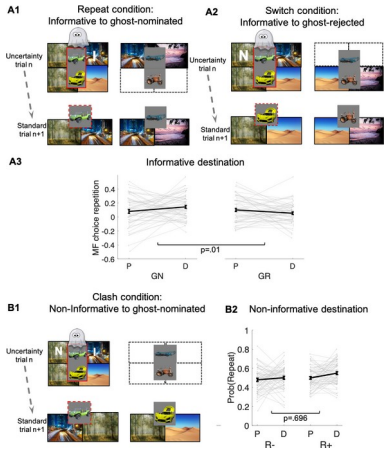
Deserno et al. (2021), Fig. 2.

Dopamine can alter retrospective credit assignment

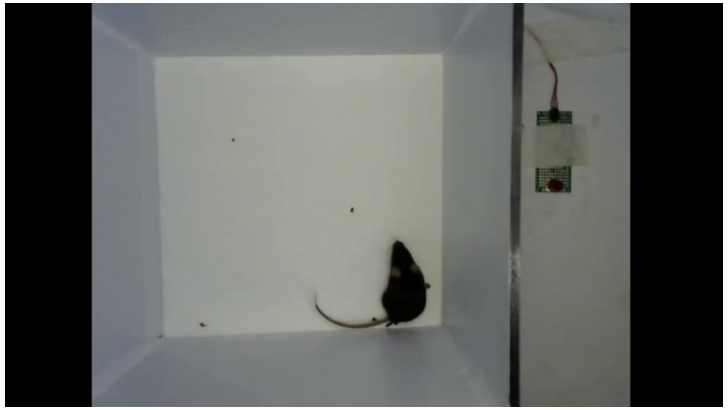
In the human credit-assignment study, levodopa increased the influence of model-based inference on model-free updating. Participants used later information about the task to assign credit to an earlier choice more strongly under the drug condition. The result concerned cooperation between learning processes rather than a uniform increase in every type of value-based decision.

The effect was distinguished from direct changes in the overall model-free and model-based contributions to choice. This separation matters because a drug can alter how information is assigned to a past action without simply making all rewards more attractive. The task and model were designed to identify that more specific change.

Retrospective inference allows information arriving after an action to refine what is learned about it. The experiment therefore extends the idea of reinforcement beyond a mechanical association with the most recent response. Knowledge of the task can guide which stored action value is updated when the consequence becomes interpretable.



Deserno et al. (2021), Fig. 4.



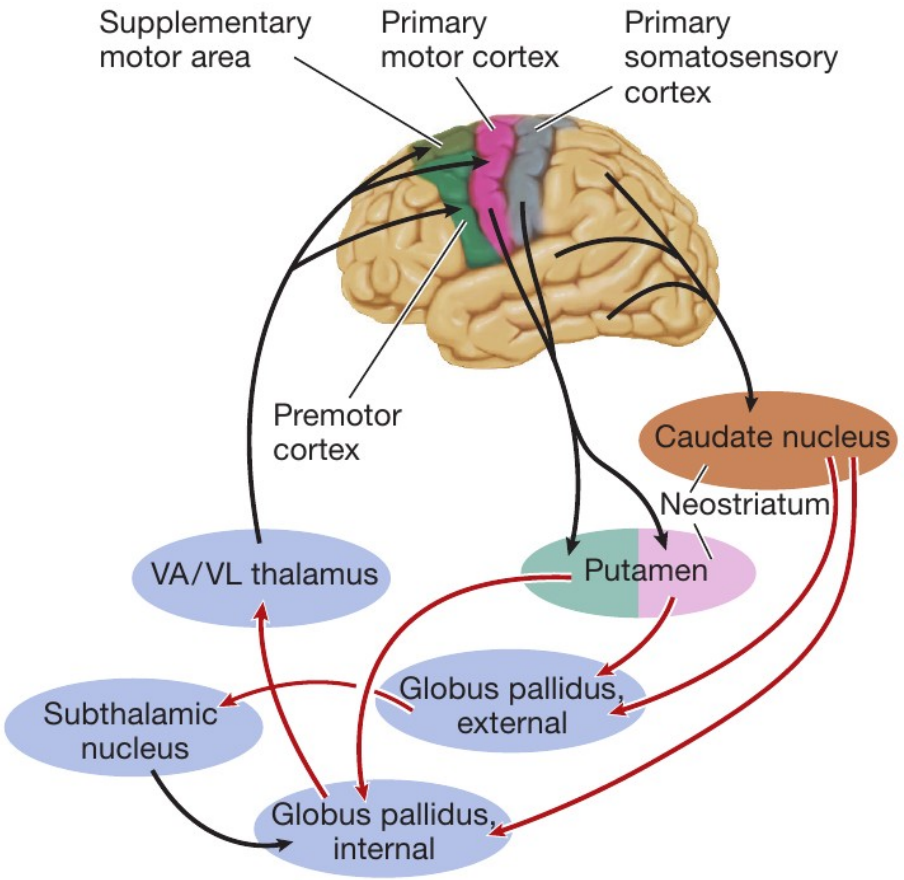
Feng et al. (2021), Video 5.

Habits reduce reliance on a freshly evaluated outcome

Goal-directed action uses knowledge of what an action produces and whether that outcome is currently valuable. Habitual action is relatively less sensitive to such immediate evaluation. Repeated practice can make responses efficient, but repetition alone does not prove that the behavior has become a habit; the relevant sensitivity must be tested.

Outcome devaluation changes how desirable a previously earned consequence is. Contingency degradation changes how reliably an action produces that consequence. If responding persists despite these changes, the pattern can indicate increased habitual control. These tests distinguish an established response tendency from continued action based on the expected value of the outcome.

Corticostriatal systems contribute to both goal-directed and habitual behavior through different interacting circuits. A person or animal can use more than one strategy within a task. Changes in practice, context, stress, and outcome information can alter their relative contributions, so habit is a property of behavioral control under tested conditions rather than a synonym for any frequent action.



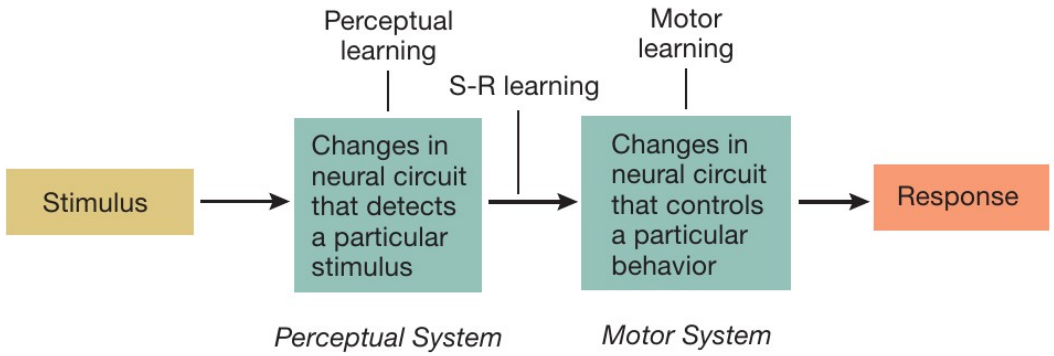
Carlson and Birkett (2022), Fig. 13.14, p. 426.

Practice changes skilled movement organization

Motor learning improves how movements are selected, coordinated, and adjusted through experience. Early practice can require deliberate attention to individual steps. With training, timing and coordination can become more reliable, and movements can be organized into useful sequences. Improvement must be distinguished from a temporary change caused by fatigue, arousal, or a more favorable test condition.

Sensory feedback provides information about the result of an attempted action. Visual, tactile, and proprioceptive signals can reveal errors in position, timing, or force. Neural circuits use these signals alongside reward and prior predictions to modify later attempts. Different tasks emphasize correcting movement errors, choosing successful actions, or acquiring a new coordination pattern.

The retention test asks whether the acquired improvement survives after practice has ended. Transfer tests ask whether it can be used under altered conditions. A movement that becomes faster during one training session may reflect a different change from a skill that remains accurate the next day or generalizes to a new object.



Carlson and Birkett (2022), Fig. 13.6, p. 420.



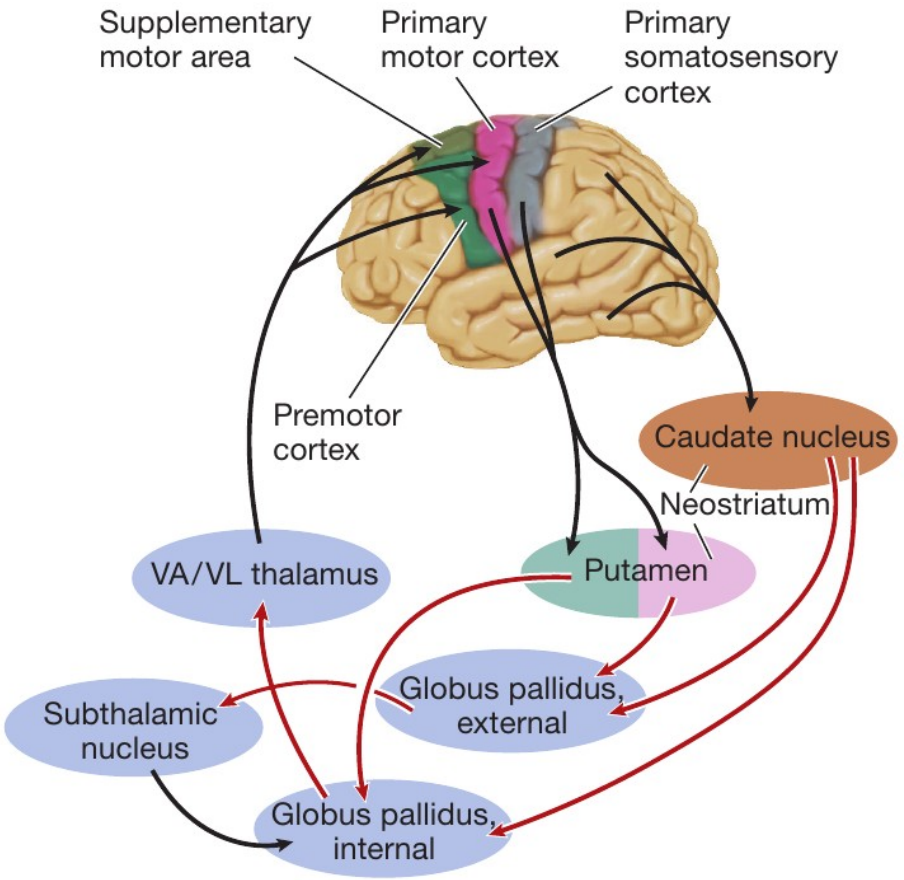
Li et al. (2024), Video 1.

Several circuits contribute to skill acquisition

Motor cortex, premotor areas, basal ganglia, and cerebellum make different contributions to learning movements. Cortex helps represent actions and their organization, corticostriatal circuits contribute to reinforcement and selection, and cerebellar pathways support prediction and error-based adjustment. Their participation changes with the task rather than following one fixed sequence for every skill.

Practice can alter the relationship among neural activity, sensory feedback, and successful movement. Some changes improve the precision of an existing pattern, while others reorganize a sequence of actions. Comparing the same movement before and after learning helps identify changes associated with skill instead of changes explained only by performing a different action.

Experimental interventions can target acquisition, retention, or execution at different times. A manipulation that disrupts performance during stimulation does not automatically show that the memory was erased. Testing the animal again after the manipulation ends helps distinguish a temporary inability to express the skill from a lasting failure to retain the improvement.



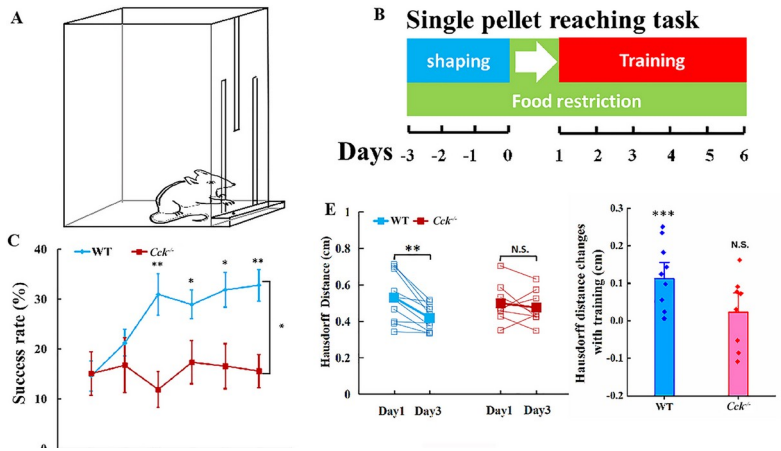
Carlson and Birkett (2022), Fig. 13.14, p. 426.

CCK signaling contributes to learning a skilled reach

A mouse study trained animals to reach through a narrow opening and retrieve a food pellet. Successful retrieval required coordinating the forelimb with the position of the pellet. The investigators compared normal animals with mice lacking CCK, following improvement across training rather than measuring only whether an untrained animal could move its paw.

Normal mice improved retrieval success with practice, whereas the CCK-deficient group remained closer to its initial performance. Blocking the CCK-B receptor also impaired learning under the tested conditions. The comparison identified a contribution of peptide signaling to skill acquisition instead of attributing the effect simply to the presence of food as a reward.

The study also examined motor-cortical plasticity and activity patterns during training. The missing peptide signal was associated with impaired refinement of the population response. Measuring behavior together with neural changes helped relate a learning deficit to the modification of motor circuitry rather than only to a description of unsuccessful reaches.



Li et al. (2024), Fig. 1.



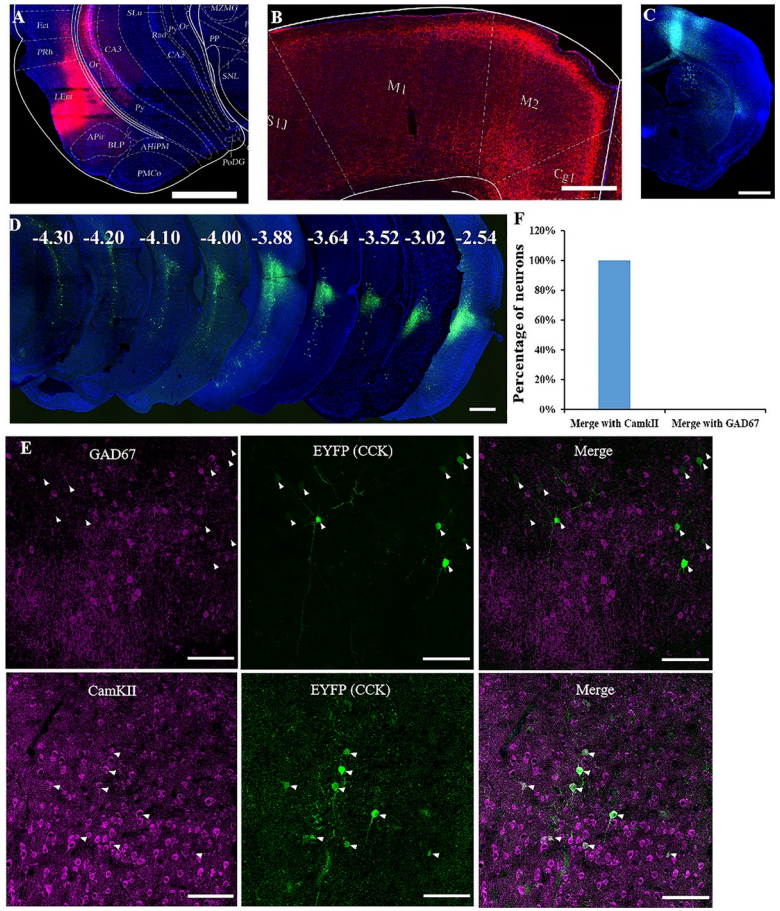
Li et al. (2024), Video 2.

Rhinal input supplies a modulatory route to motor cortex

The skilled-reaching study traced projections from CCK-expressing neurons in rhinal cortex to motor cortex. This provided an anatomical route by which a peptide signal outside the motor area could influence its plasticity. The labeled axons identified a candidate source, while later intervention tested the contribution of the projecting population.

Chemogenetic inhibition of the motor-projecting rhinal neurons reduced skill learning. Supplying a CCK agonist could rescue important aspects of the learning deficit in CCK-deficient mice. These manipulations tested the source population and the availability of peptide signaling through complementary approaches rather than relying on anatomy alone.

CCK altered conditions for plasticity without specifying the complete motor sequence in the peptide molecule itself. Information about the target, paw position, and successful retrieval still came through active sensory and motor pathways. The modulatory route influenced how those pathways changed with practice, helping distinguish a permissive signal from the detailed pattern being learned.



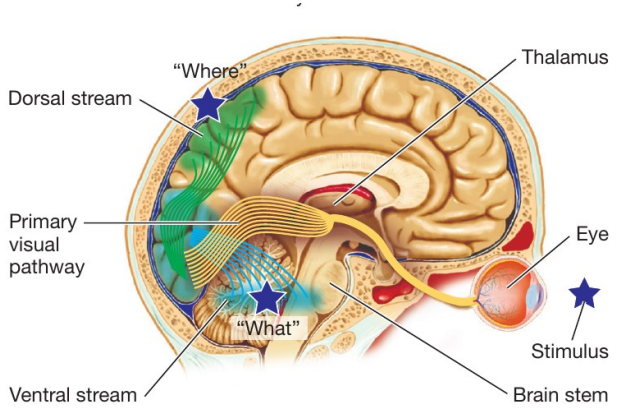
Li et al. (2024), Fig. 4.

Perceptual learning changes stimulus discrimination

Perceptual learning changes how effectively an organism detects, distinguishes, or recognizes sensory information through experience. Repeated exposure can improve discrimination of a visual orientation, an unfamiliar sound pattern, or a tactile feature. The change can involve sensory representations, attention, decision processes, and the readout of relevant signals rather than a single universal storage site.

Sensory association cortex participates in recognizing complex patterns and retrieving information about them. Different cortical populations represent different features, such as object form or motion. A learning effect can therefore be specific to the trained feature or sensory conditions, depending on which representation and decision process were modified during practice.

Improved recognition need not include a conscious recollection of each earlier exposure. A person with impaired event memory may still show perceptual learning on some tasks. This dissociation separates familiarity or discrimination produced by experience from the ability to describe when and where that experience occurred.



Carlson and Birkett (2022), Fig. 13.18, p. 432.



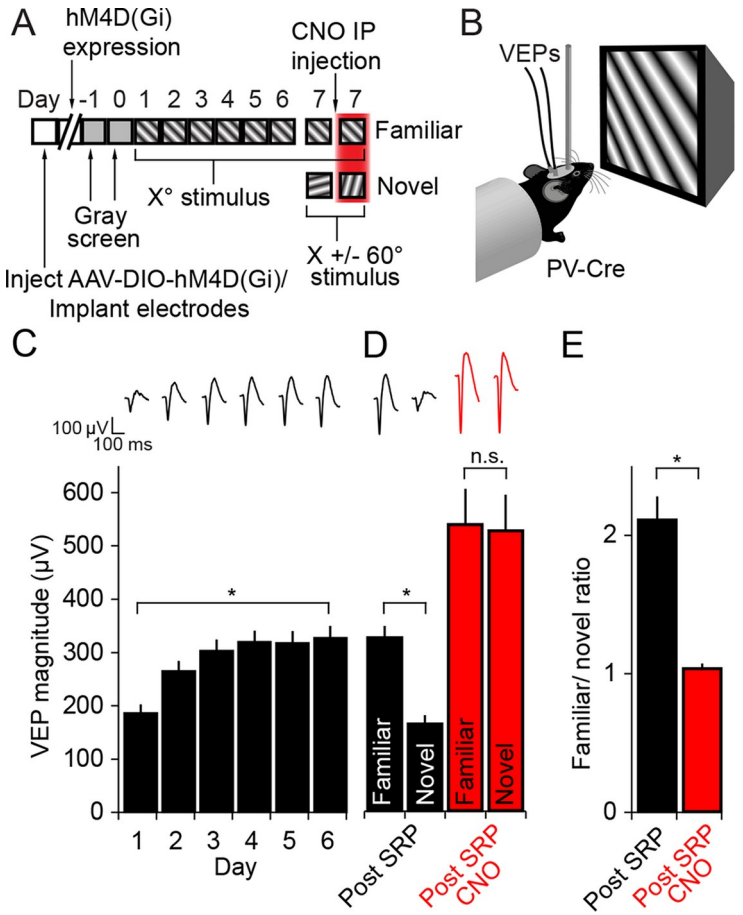
Li et al. (2024), Video 3.

Visual familiarity depends on inhibitory cortical cells

A mouse study repeatedly presented a particular visual orientation and measured stimulus-selective response potentiation in primary visual cortex. The response to the experienced stimulus increased with exposure. The investigators targeted parvalbumin-expressing inhibitory neurons to test whether these cells contributed to the expression of this experience-dependent change.

Temporarily inhibiting the parvalbumin population disrupted expression of the potentiated response. The manipulation acted within a local inhibitory circuit rather than directly removing the visual stimulus. The result identified a contribution of inhibition to the expression of a learned sensory response, showing that greater cortical responsiveness need not be explained only by stronger excitation.

The study compared this process with ocular-dominance plasticity after altered input from one eye. Although both forms of experience could change cortical responses, they differed in their dependence on the targeted inhibitory cells. Similar-looking physiological outcomes can therefore arise through distinguishable circuit mechanisms.



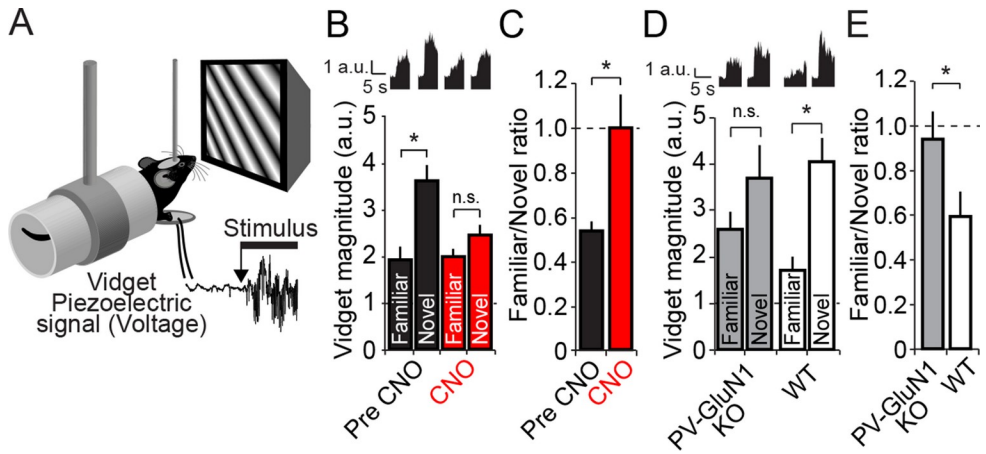
Kaplan et al. (2016), Fig. 3.

Familiarity behavior can be separated from visual input

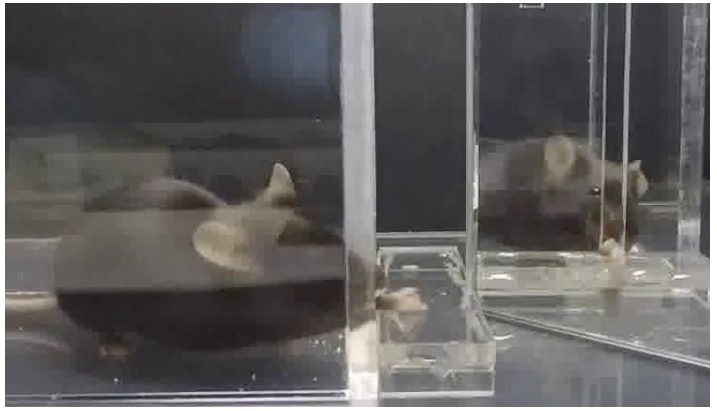
The visual-plasticity study also measured behavioral responses to familiar and novel oriented patterns. Normal mice distinguished the experienced stimulus from a new one. This provided a behavioral counterpart to the physiological change recorded in visual cortex, allowing the experiment to test whether the altered response mattered beyond the electrode signal.

Manipulating parvalbumin-cell function and their NMDA receptors disrupted important aspects of familiarity recognition. The relevant receptor contribution was located within the inhibitory population, not simply assumed to reside only in excitatory pyramidal neurons. The comparison linked a defined receiving mechanism to the expression of experience-dependent cortical and behavioral responses.

A novel visual pattern still supplies sensory input even when familiarity discrimination is impaired. The task therefore separates detecting the display from responding differently because of prior experience. This distinction is essential when interpreting a learning deficit: a loss of recognition must be evaluated alongside the ability to see and respond to the stimulus at all.



Kaplan et al. (2016), Fig. 8.



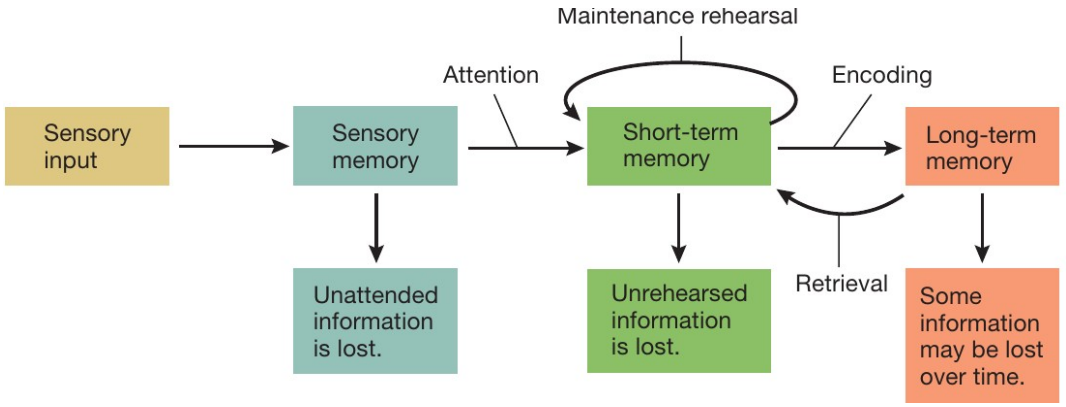
Li et al. (2024), Video 4.

Short-term retention keeps selected information available

Sensory memory briefly preserves information after a stimulus disappears. A smaller selected portion can remain available in short-term or working memory. These descriptions concern different operations and durations, rather than physical boxes through which every experience must pass in exactly the same way. Attention and task requirements influence which information remains useful.

Rehearsal can maintain some information, such as silently repeating a short sequence of digits or sounds. Other forms of retention depend on sensory representations, spatial attention, or an active decision variable. The available capacity is limited, and interference can disrupt performance before a long-term memory has been established.

A delay task measures more than retention alone. The animal or person must encode the initial information, maintain or reconstruct it, and select a response at the test. Manipulations applied at different stages help identify whether an error arose during the original perception, the intervening delay, or the later use of the retained information.



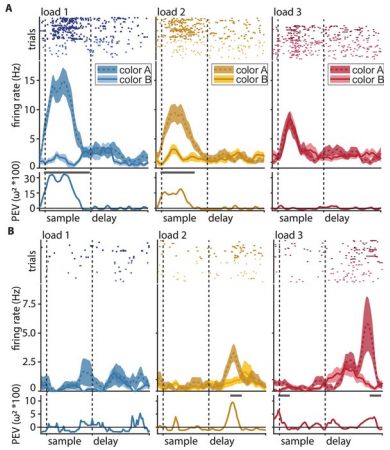
Carlson and Birkett (2022), Fig. 13.7, p. 421.

Memory load changes neuronal information in crows

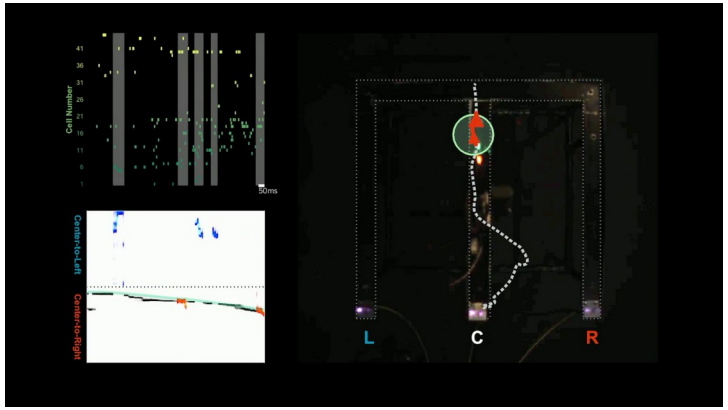
A comparative study used a color-change detection task to test working memory in crows and monkeys. Subjects briefly viewed several colored items and later indicated a change. Increasing the number of items increased memory load, allowing performance and neural information to be compared as more material had to remain available.

Extracellular recordings in the crow nidopallium caudolaterale measured how neuronal responses represented the colors. This associative forebrain region differs anatomically from mammalian prefrontal neocortex. The experiment therefore tested whether similar computational constraints could appear in different neural architectures rather than assuming that a mammalian cortical layout was required.

For many neurons, information about the relevant color declined as load increased, while some cells showed a different pattern. The population did not behave like a collection of identical fixed-capacity slots. Measuring both behavior and individual neural responses helped identify how greater demand changed the quality of the retained representation.



Hahn et al. (2021), Fig. 2.



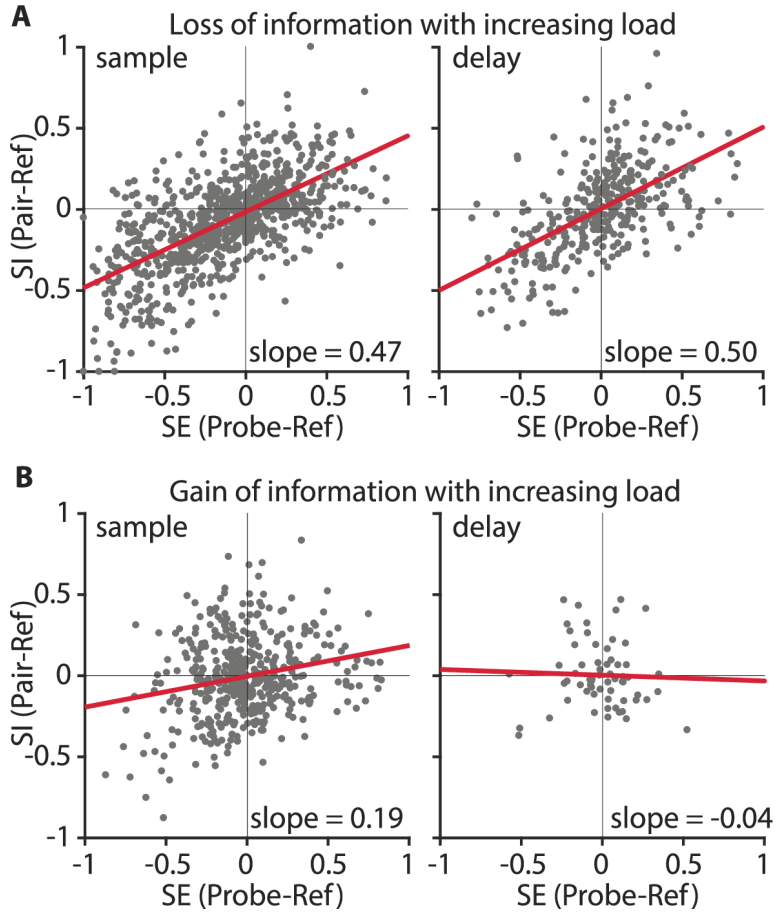
Tang et al. (2021), Video 1.

Normalization can limit a retained representation

The crow-and-monkey study evaluated divisive normalization as an explanation of load effects. In this computational description, a neuron's response is scaled relative to activity representing other information. Adding items can therefore reduce the distinctiveness of one item's representation without requiring the neuron to stop firing completely.

Neurons losing information at higher loads showed response patterns consistent with normalization-like regularization. Other neurons did not follow the same pattern. The result supported a shared computational principle across the tested bird and primate systems while preserving differences among cells within each recorded population.

A computational description specifies a relation among measured responses; it does not by itself identify every synapse that implements the operation. The study compared model predictions with physiology and behavior. Similar load-dependent computations could emerge in distinct anatomical structures, allowing comparative neuroscience to test function without requiring identical brain geometry.



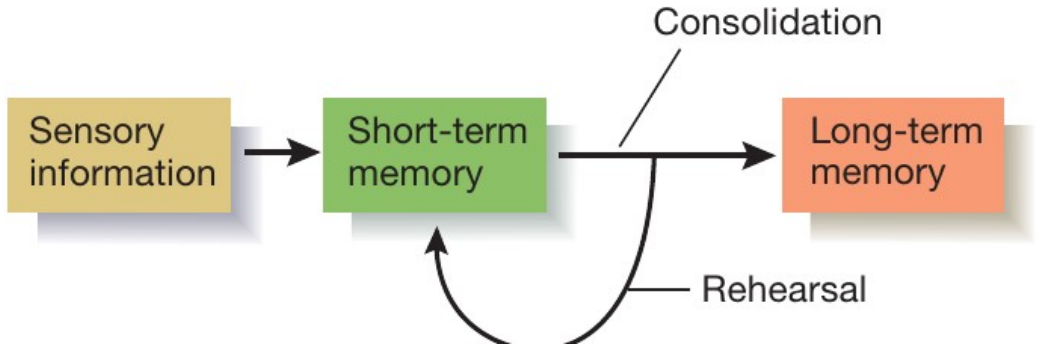
Hahn et al. (2021), Fig. 5.

Working memory maintains information for current action

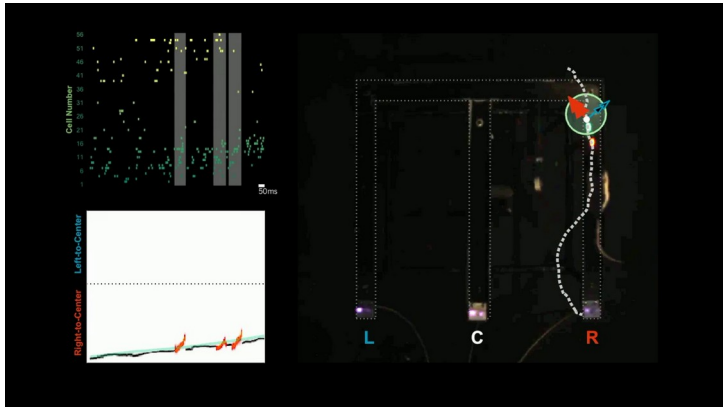
Working memory refers to maintaining and using information needed for an ongoing task. It can include holding a sound sequence, comparing two objects, or keeping a planned movement available. Prefrontal and sensory-associated regions contribute through interacting activity patterns, with different demands recruiting different networks rather than one isolated memory center.

Persistent firing can carry information across a delay, but useful information can also be represented through changing population patterns and other circuit states. A stable behavioral memory does not require every contributing neuron to fire at a constant rate. The experiment must test whether the relevant information remains available despite the changing activity.

Load, distraction, and the strategy used to summarize information affect performance. Retaining several separate observations differs from retaining one value computed from them. The chosen strategy can therefore change the apparent capacity limit, even when the person sees the same original set of stimuli.



Carlson and Birkett (2022), Fig. 13.8, p. 421.



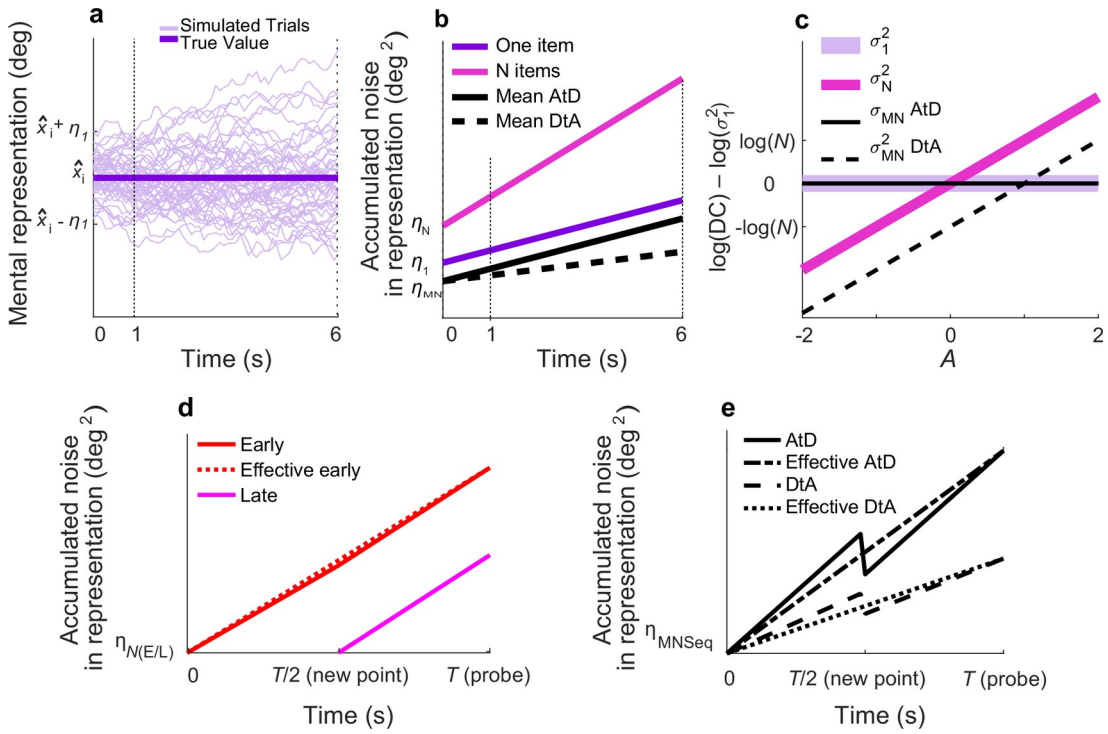
Tang et al. (2021), Video 2.

Strategy changes what working memory stores

A human experiment asked participants to report the average location of multiple visual targets. The task could be approached by calculating the average immediately or by retaining separate target locations and computing the answer later. These strategies place different demands on memory despite beginning with the same kind of sensory observations.

The investigators built models of the noise and decay expected under each strategy. Storing a single summary value reduces the number of separate items that must be retained. Storing several locations preserves more detail but allows additional degradation or interference before the eventual report is produced.

Behavioral tests compared the predicted error patterns with actual reports across delays and item loads. The purpose of the model was to distinguish possible information-processing routes, not to present simulated performance as a biological observation. The measured errors supplied evidence about which representation participants were likely maintaining under a particular task condition.



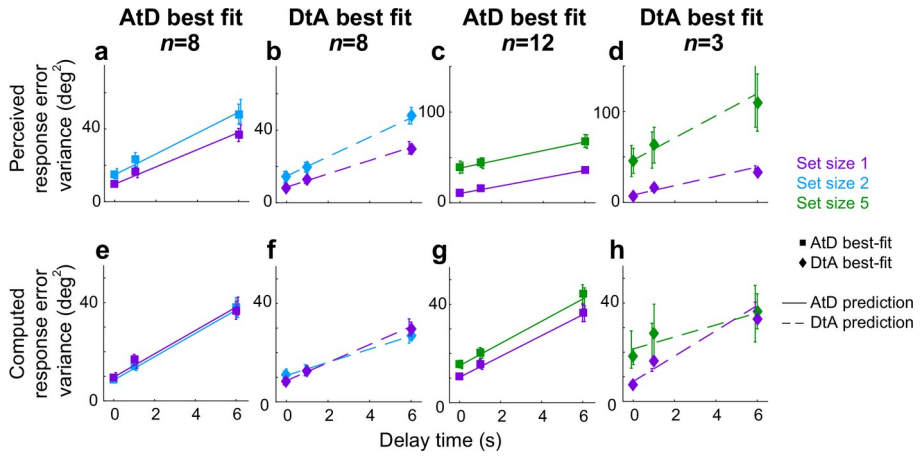
Schapiro et al. (2022), Fig. 2.

Memory errors depend on when information is combined

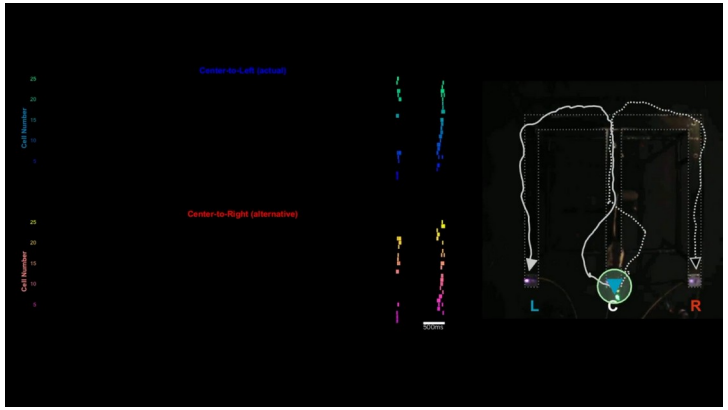
The visual-average study found that the effects of delay and capacity depended on strategy. When information could be summarized promptly, memory could preserve one decision variable. When several values had to remain separate until the report, errors developed differently as time and load increased.

The comparison shows why a working-memory capacity estimate depends on what counts as an item in the task. Several observations can be compressed into one useful value, while a later requirement may demand their individual identities. The same sensory sequence can therefore produce different memory demands according to the computation performed during encoding.

These results also separate a limit on retaining information from a limit on combining it at the end. A report can be inaccurate because the original values degraded or because their final integration was noisy. Manipulating the required strategy helps identify which operation contributed to the observed error.



Schapiro et al. (2022), Fig. 5.



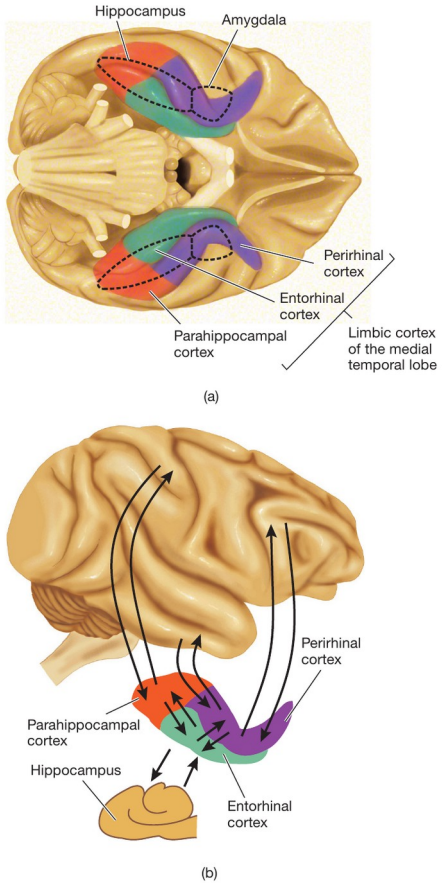
Tang et al. (2021), Video 3.

Relational memory represents events in their context

Relational learning represents how pieces of information fit together, such as an object's location, the order of events, or who participated in an experience. An episodic recollection uses relationships among these elements to specify a particular event. Recognizing an object as familiar can occur without reconstructing all of its original context.

The hippocampal formation receives processed sensory and associative information through medial-temporal cortical pathways. Its circuitry can combine inputs that originated in different sensory systems. This arrangement allows a location, sound, object, and action to participate in a common event representation rather than remaining only separate sensory traces.

Relational knowledge can guide flexible action. Knowing that food is hidden at a site can lead an animal to visit it when retrieving and avoid adding another item there when storing food. The retained content can remain similar while the action it supports changes with the current goal.



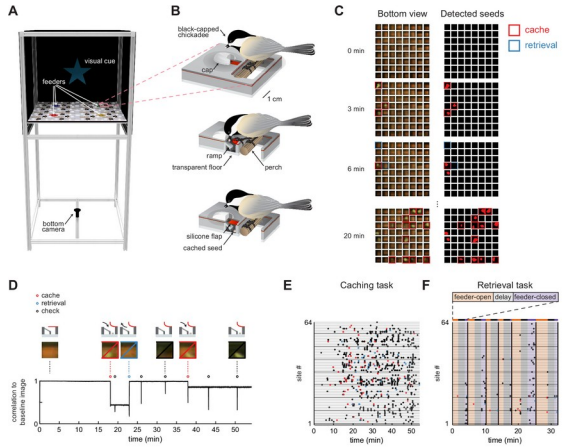
Carlson and Birkett (2022), Fig. 13.20, p. 434.

Food-caching birds reveal precise spatial memory use

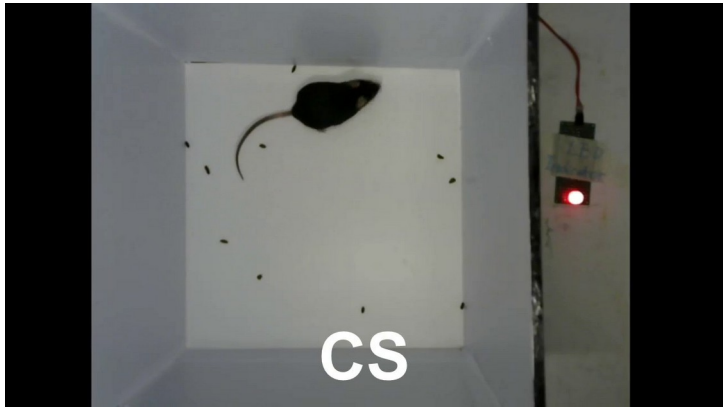
A chickadee study used an arena with many potential caching sites. The apparatus tracked where birds stored food, which sites they investigated, and where they later retrieved it. This natural behavior provided a way to study spatial memory while also measuring the movements and site preferences that could influence navigation independently of memory.

The birds used precise information about occupied sites and sites already checked. They also showed spatial biases and opportunistic foraging. The researchers therefore compared models containing both memory-related and memory-independent factors, rather than attributing every visit to a successful recollection of hidden food.

Automated tracking distinguished placing a cache, retrieving it, and inspecting an empty site. These actions carry different information about what the bird knows. A route through the arena alone cannot establish whether an animal remembered a particular cache, while the sequence of site interactions can test that possibility more directly.



Applegate et al. (2022), Fig. 1.



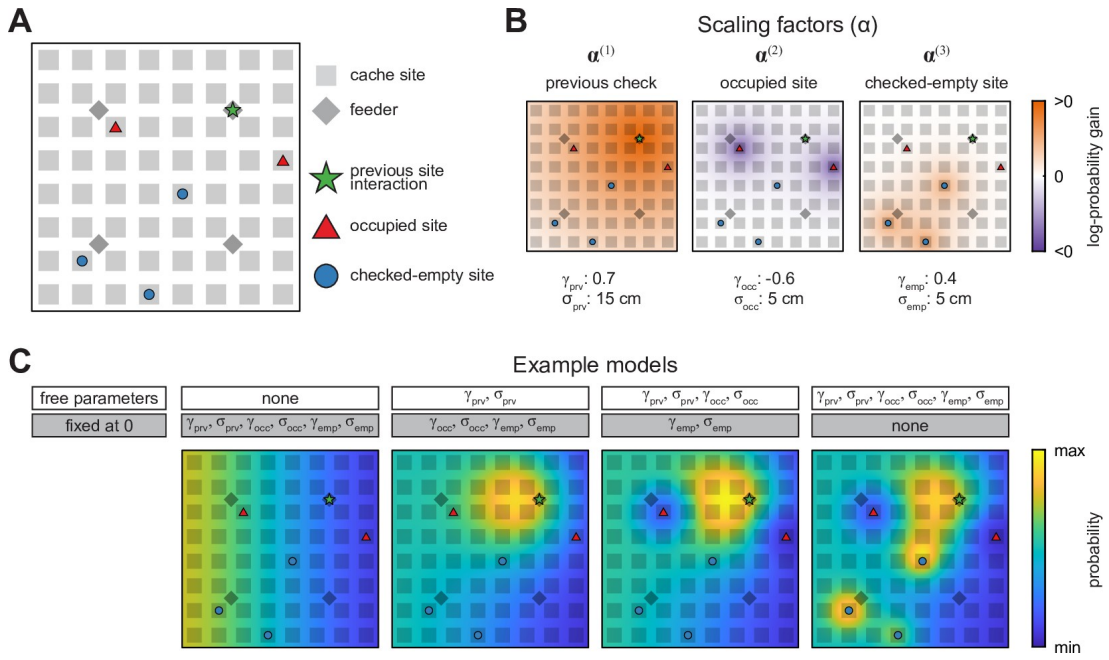
Feng et al. (2021), Video 9.

One spatial memory can support opposite choices

The chickadee study tested whether information about site contents was used differently during caching and retrieval. During caching, birds tended to avoid sites that already contained food. During retrieval, they preferentially accessed occupied sites. The remembered content therefore influenced action according to the current behavioral goal.

Probabilistic models separated these memory effects from distance, spatial preference, and other navigation factors. A single retained representation of site contents could be read out differently in the two contexts. The animal did not need one fixed stimulus–response rule in which an occupied site always caused either approach or avoidance.

This flexibility distinguishes relational use of information from simply repeating a previously rewarded route. The same location can become a different choice when the purpose of the visit changes. Measuring both contexts allowed the experiment to identify how retained spatial content was combined with the current task.



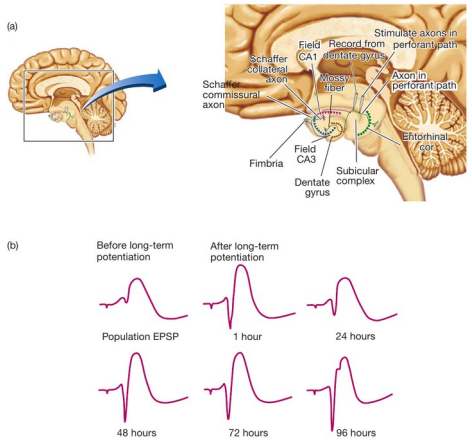
Applegate et al. (2022), Fig. 3.

Hippocampal subfields transform incoming information

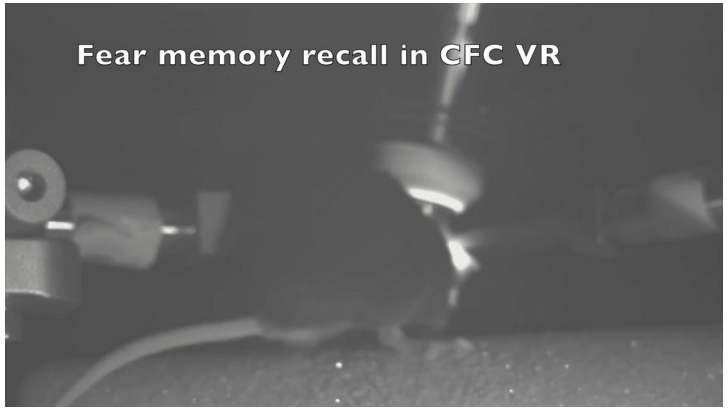
The hippocampal formation includes the dentate gyrus, hippocampal fields such as CA3 and CA1, and the subiculum. Entorhinal cortex provides major input, with pathways reaching different components. Connections within the formation transform and distribute the incoming activity before output returns to cortical and subcortical targets.

Dentate granule cells provide strong input to CA3 through mossy fibers. CA3 pyramidal neurons also communicate through recurrent connections and send output toward CA1. CA1 receives both processed hippocampal input and additional cortical information. These pathways create several opportunities for patterns to be separated, associated, reactivated, or compared.

Inhibitory interneurons regulate which cells become active and when they fire. A memory-related ensemble therefore reflects coordinated excitation and inhibition rather than a group of permanently activated cells. The same anatomical circuit can participate in different experiences by recruiting different combinations of cells and altering their later responsiveness.



Carlson and Birkett (2022), Fig. 13.30, p. 448.



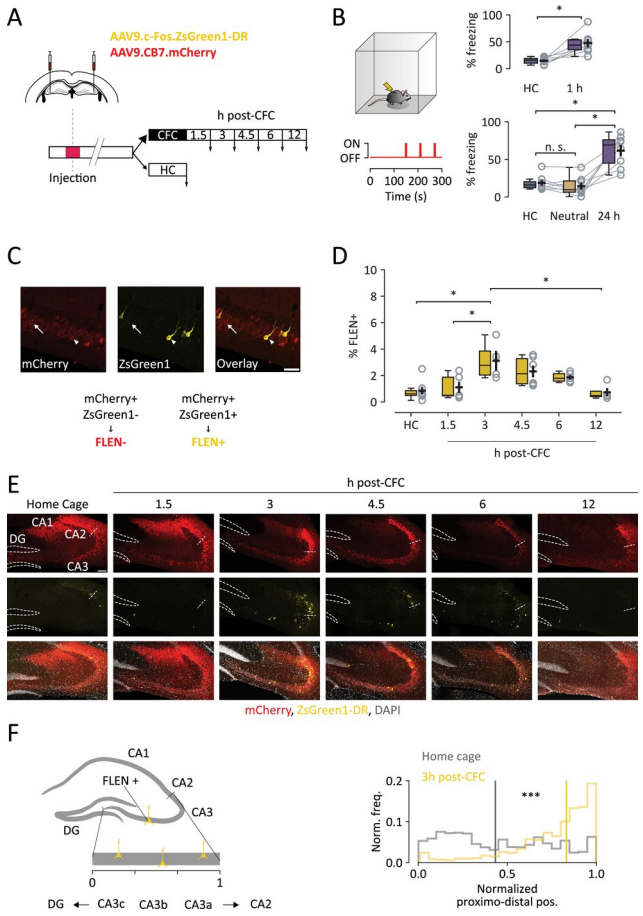
Krishnan et al. (2025), Video 3.

Activity labels reveal early changes in CA3 ensembles

A recent mouse study developed fast labeling of neurons activated during a salient experience. The FLEN tool used activity-dependent gene regulation to produce a short-lived fluorescent marker. This allowed investigators to identify CA3 neurons recruited within hours of a one-trial experience, rather than waiting only for a label that appeared much later.

The marker was compared with another activity-labeling system used at a later interval. Labeled cells could then be studied with electrical recordings from brain tissue. The experiment asked how the properties of an experience-associated population changed across the early consolidation period, using the label to identify which cells had been recruited.

Activity-dependent labeling identifies cells that met the tool's expression conditions during a selected window. It does not display a complete remembered scene inside the microscope. Comparing labeled and unlabeled cells tests whether recruitment was associated with measurable differences in excitability or synaptic input under the specified recording conditions.



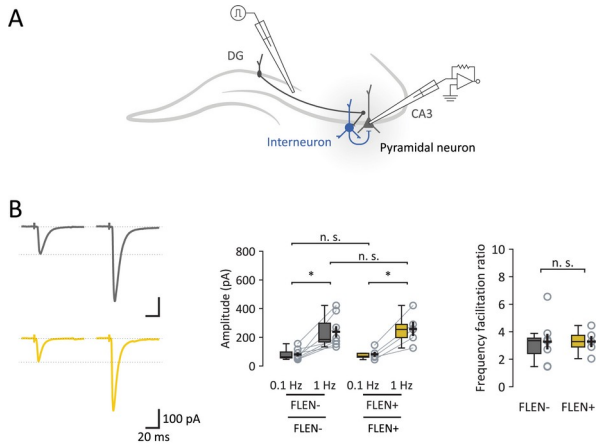
Cupolillo et al. (2025), Fig. 2.

Engram-cell properties evolve after a single experience

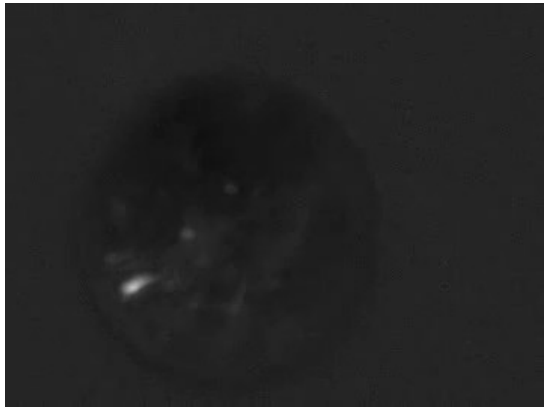
The CA3-labeling study compared intrinsic excitability and synaptic properties at different post-experience intervals. Intrinsic excitability describes how readily a cell produces spikes in response to input. Synaptic measurements describe communication arriving through connections, allowing the investigators to distinguish a change within the cell from a change in the input it receives.

Early labeled neurons already showed increased excitatory input, while increased intrinsic excitability was more evident in the later labeled population. The properties did not all change simultaneously. An experience-associated ensemble could therefore evolve through successive cellular modifications during the first day rather than acquiring one complete permanent state at the instant of learning.

Recordings also examined mossy-fiber input and feedforward inhibition in CA3. These measurements placed the labeled cells within an active circuit instead of treating them as isolated memory containers. The timing of the label and recording determined which phase of the developing neural change the experiment could observe.



Cupolillo et al. (2025), Fig. 5.



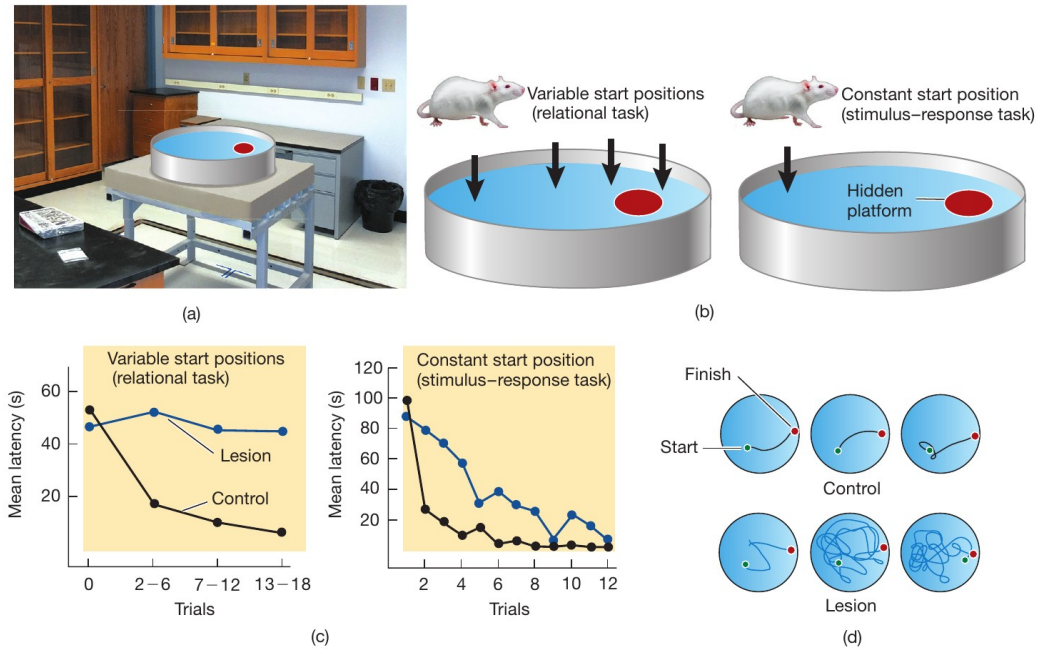
Twarkowski et al. (2022), Video 1.

Place cells encode location in relation to experience

Place cells are hippocampal neurons whose activity is elevated in particular parts of an environment. The region associated with a cell's response is called its place field. Across a population, different cells respond at different positions, creating a pattern that can distinguish locations during navigation.

Place responses also depend on context, task demands, and experience. A cell can change its field when the environment or its meaning changes, a process called remapping. The same physical location can therefore be represented differently after a meaningful event, rather than functioning as an unchanging coordinate stored by one dedicated neuron.

Experiments compare population activity while tracking position and behavior to separate location from speed, direction, reward, or an impending choice. Spatial selectivity alone does not establish every memory function of a cell. Its response must be interpreted within the route, experience, and behavioral condition being tested.



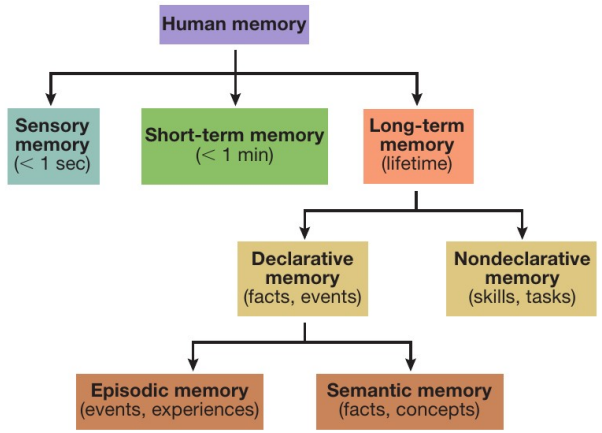
Carlson and Birkett (2022), Fig. 13.23, p. 438.

Episodic and semantic memories retain different information

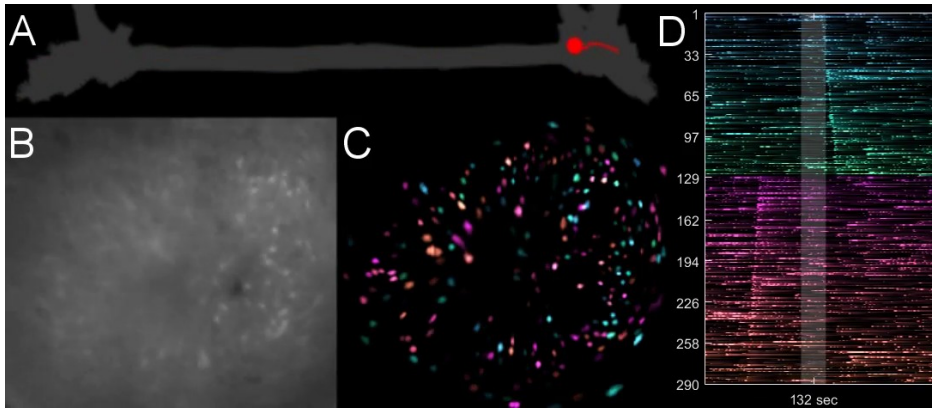
Episodic memory concerns an event situated in a particular context, including what happened and aspects of where or when it occurred. Semantic memory concerns knowledge, such as facts and concepts, that need not include the original learning episode. Knowing that a city exists differs from recalling a particular visit to it.

Both forms depend on distributed neural representations, and their development can interact. Repeated experiences can contribute to knowledge that becomes less tied to one occasion. Conversely, semantic knowledge can help organize the interpretation and later retrieval of a new episode. The distinction identifies retained content rather than two completely isolated brain systems.

The hippocampus is particularly important for relationally detailed episodic processing, while cortical networks represent many features of knowledge and perception. The degree of hippocampal dependence can vary with the detail and task being tested. A broad statement that an old memory has moved entirely out of the hippocampus can therefore miss the distinction between general knowledge and vivid event reconstruction.



Carlson and Birkett (2022), Fig. 13.9, p. 422.



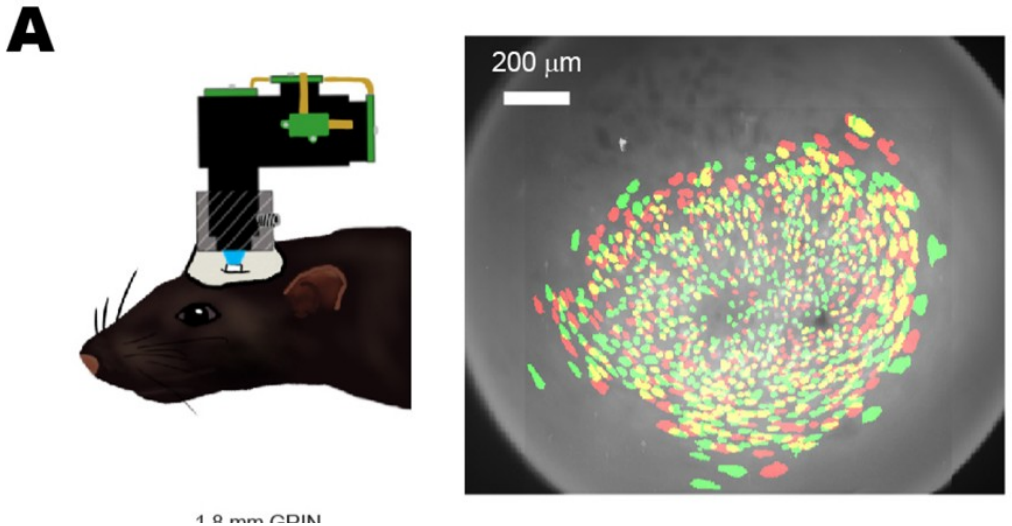
Blair et al. (2023), Video 1.

Remembered aversive events alter place-cell maps

A rat study recorded calcium activity from hippocampal CA1 neurons while animals chose between a short and a long path to food. Rats initially preferred the short path. An aversive event delivered on that route then produced avoidance, allowing the researchers to compare neural maps before and after the experience.

The population representation changed more after an event that was remembered than after a comparable event under an amnesia-producing drug condition. The researchers also compared an ordinary barrier. These conditions helped distinguish remapping associated with retained aversive experience from any change caused simply by encountering an obstacle or moving differently.

Population comparisons were made across sessions as well as within sessions, while cells were tracked over time. The experiment therefore examined the persistence of a changed representation. A transient response at the moment of the event and a reorganized map retained later were different outcomes measured within the same preparation.



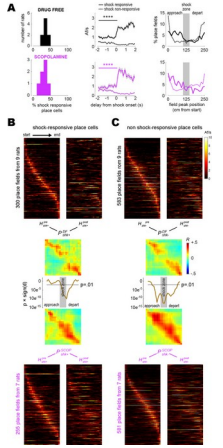
Blair et al. (2023), Fig. 2.

Shock responses do not fully explain later remapping

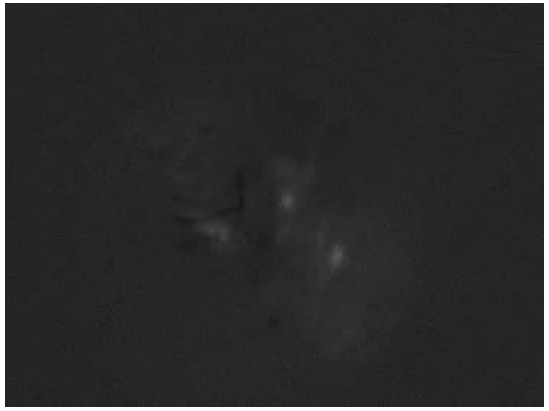
The place-remapping study separately measured the immediate calcium responses of place cells to the aversive event. Some neurons responded strongly at that moment, while others did not. The investigators asked whether only the directly responsive cells later changed their spatial fields, which would suggest a simple explanation based on immediate activation.

Remapping was not restricted to the neurons showing the strongest immediate response. The drug condition could preserve spatial tuning and responses to the event while weakening later avoidance and reducing remapping. Sensory detection and a short-lived neural reaction therefore did not guarantee the same persistent change in the population code.

The changed fields did not simply accumulate at the aversive location or all move away from it. A remembered event reorganized the broader representation. This separates a local response to stimulation from the later way an environment was encoded across the population after learning.



Blair et al. (2023), Fig. 4.



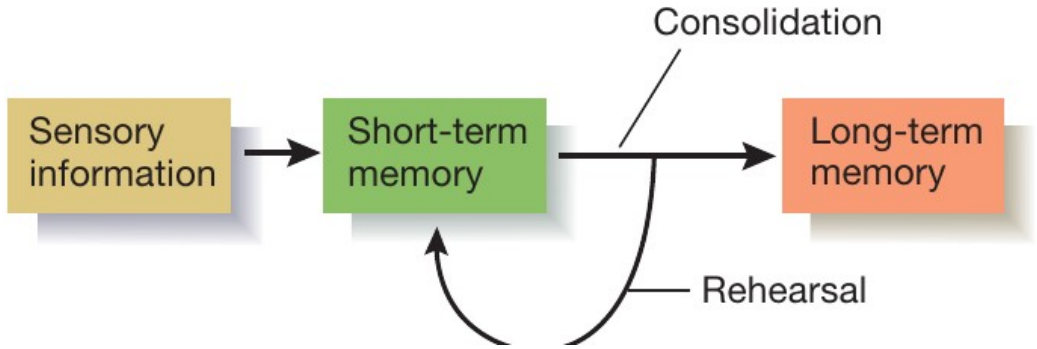
Twarkowski et al. (2022), Video 2.

Consolidation changes how memory is maintained over time

Consolidation includes cellular stabilization and changes in coordination across brain regions after learning. Some processes modify recently active synapses, while others reorganize how hippocampal and cortical populations contribute to later retrieval. These levels overlap but should not be treated as one event that occurs at a single fixed delay after every experience.

The hippocampus can help coordinate representations distributed across cortex. Repeated reactivation can change the effectiveness and organization of these networks. The resulting memory may retain detailed context or become more generalized, depending on experience and the information required at retrieval. Older age alone does not specify which features remain dependent on the hippocampus.

Experimental tests manipulate activity after encoding and measure later behavior or ensemble organization. This timing helps distinguish a role in initial perception from a role in subsequent stabilization. Sleep and quiet waking provide opportunities to study reactivation when the original stimulus is absent and the animal is not repeating the training behavior.



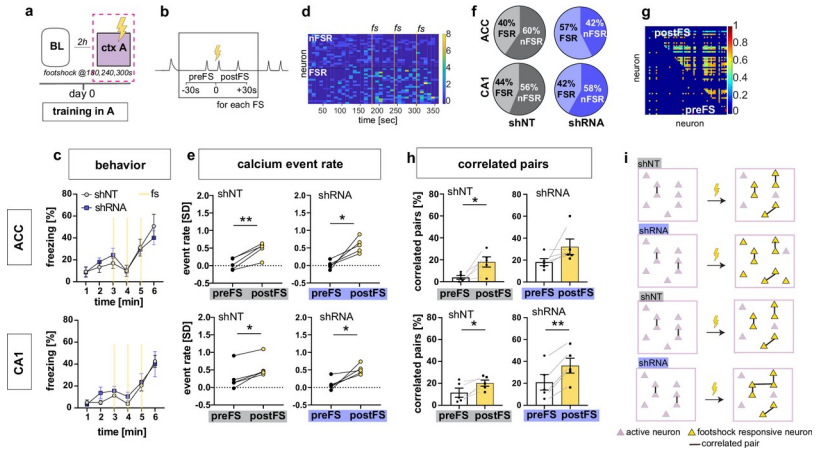
Carlson and Birkett (2022), Fig. 13.8, p. 421.

Feedforward inhibition changes later ensemble organization

A mouse study manipulated a dentate-gyrus-to-CA3 circuit that recruits inhibitory interneurons. Feedforward inhibition occurs when incoming activity activates an inhibitory cell that restrains downstream principal neurons. The investigators selectively increased this recruitment and followed activity in hippocampal CA1 and anterior cingulate cortex during contextual learning.

Increasing the inhibitory route changed neuronal responses during encoding and improved aspects of context-associated ensemble organization. More inhibition did not simply eliminate memory-related activity. By controlling recruitment and timing, the circuit altered which populations responded to a particular context and how those patterns were maintained over later sessions.

The study used longitudinal calcium imaging to follow ensembles rather than relying only on an endpoint count of activated cells. This allowed changes in context specificity and persistence to be evaluated over time. A circuit manipulation could therefore alter the evolution of a distributed representation after the initial experience.



Twarkowski et al. (2022), Fig. 2.



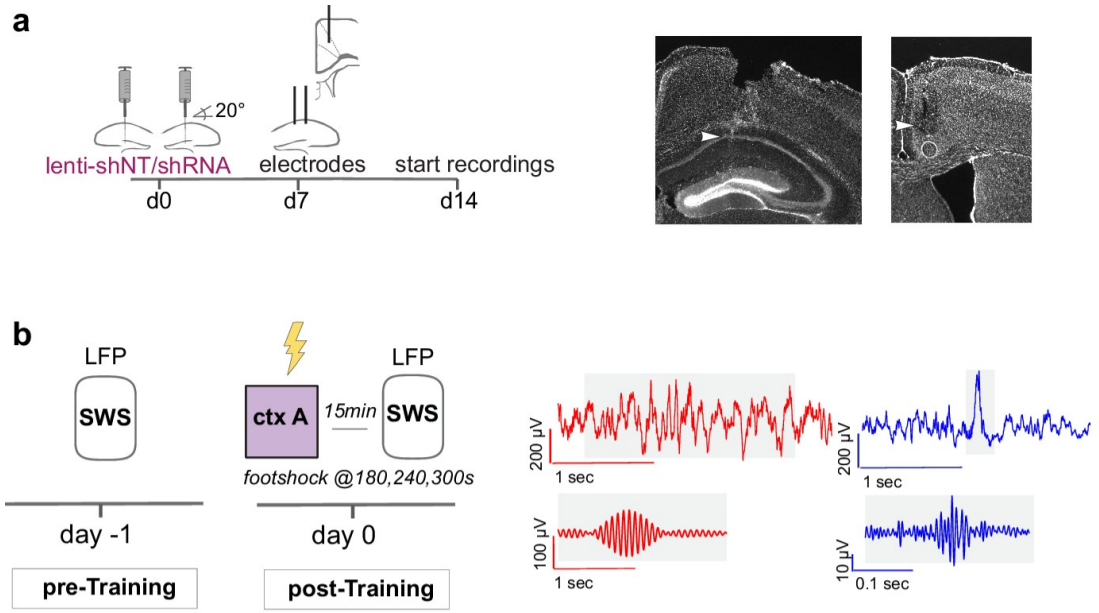
Twarkowski et al. (2022), Video 3.

Inhibition changes ripple–spindle coordination

The dentate-gyrus–CA3 study also recorded local field potentials, electrical signals reflecting coordinated population activity. Hippocampal sharp-wave ripples and cortical spindles are brief activity patterns associated with offline processing. Their temporal alignment provides a measure of coordination between regions rather than a direct image of the content being remembered.

Increasing feedforward inhibition in the targeted circuit enhanced coupling between CA1 ripples and anterior cingulate spindles. The manipulation also improved long-term contextual memory and aspects of ensemble specificity. These observations identified a candidate network process through which a local inhibitory change could influence communication between hippocampus and cortex.

Temporal coupling alone does not prove that one region transferred a complete memory to the other. The experiment combined the timing measure with circuit intervention, neural-population tracking, and behavior. Together these measurements constrained the mechanism to a defined hippocampal pathway and its effects on later distributed activity.



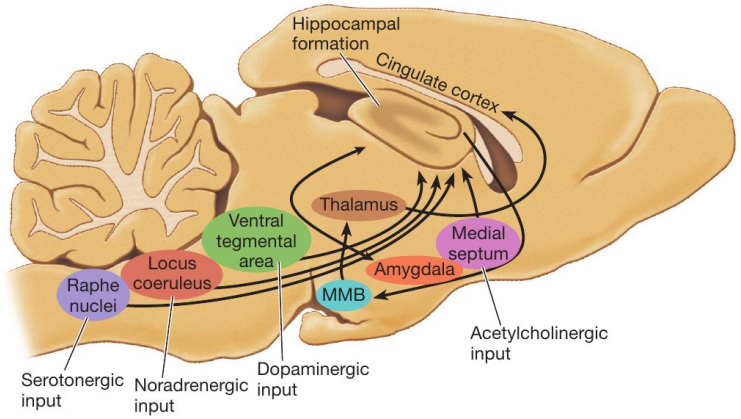
Twarkowski et al. (2022), Fig. 5.

Sleep and quiet waking permit offline reactivation

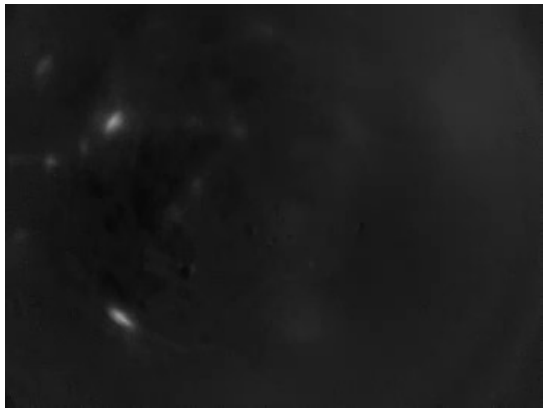
Neural patterns related to prior experience can recur when an animal is no longer performing the original task. Reactivation can occur during sleep and quiet waking, including periods containing hippocampal sharp-wave ripples. Replay refers more specifically to recurrence of an ordered pattern, such as a compressed sequence of place-cell activity associated with an earlier route.

These events allow researchers to study memory-related activity without presenting the original stimulus again. The timing of a ripple can be compared with cortical activity, behavior, and later retention. An association with improved memory is different from demonstrating necessity, which requires interventions that selectively disrupt the relevant events and measure the consequence.

Sleep contains several changing neural states, and memory-related effects depend on the task and recorded process. A spindle, ripple, or slow oscillation is a physiological pattern rather than a direct report of subjective remembering. Their coordination can provide a window into how recently modified circuits communicate during periods without active training.



Carlson and Birkett (2022), Fig. 13.21, p. 435.



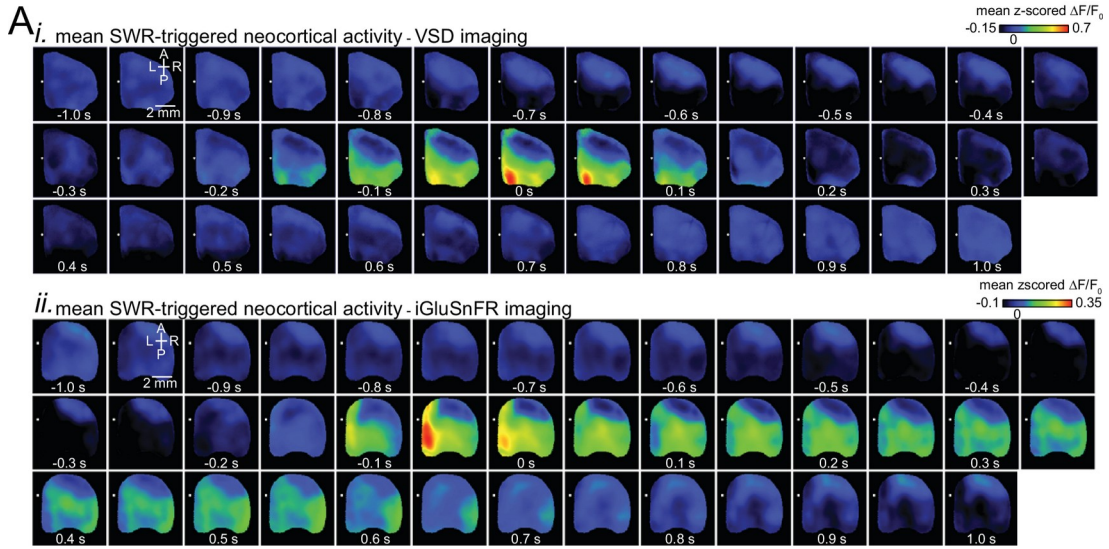
Twarkowski et al. (2022), Video 4.

Cortical regions differ in activity around ripples

A mouse study combined hippocampal recordings with wide-field measures of cortical voltage or extracellular glutamate transients. The researchers aligned activity across regions to hippocampal sharp-wave ripples. This approach measured when cortical activity changed relative to the event, rather than assuming that every ripple initiated a uniform wave traveling outward from the hippocampus.

Retrosplenial cortex showed especially large ripple-related modulation, while other regions had different patterns. The cortical response therefore had spatial structure rather than a single whole-cortex signal. Comparing the regions helped identify where communication was strongest under the measured conditions and how the activity evolved around the hippocampal event.

Voltage and glutamate sensors report different aspects of neural function, and their measurements were interpreted accordingly. Aligning their signals to the same physiological event supported a timing analysis across the network. It did not require the assumption that fluorescence directly displayed a recalled route or a complete transferred memory.



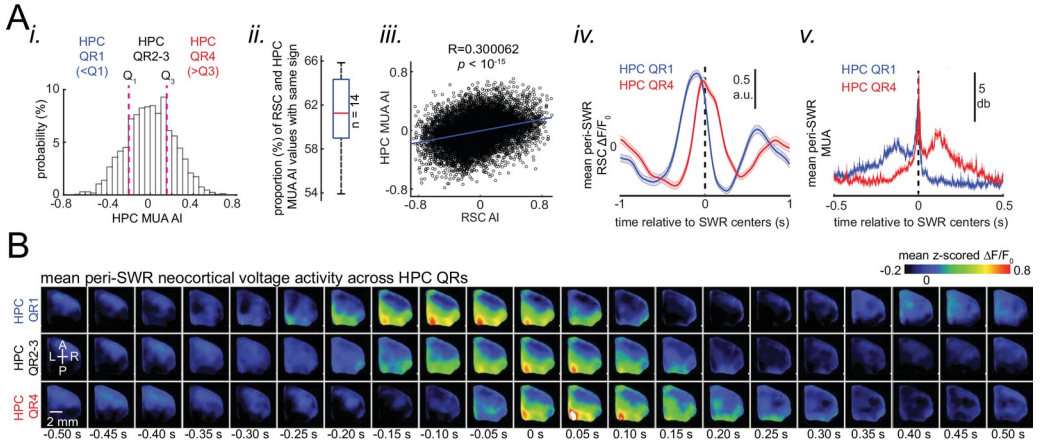
Karimi Abadchi et al. (2020), Fig. 2.

Cortical activation can precede a hippocampal ripple

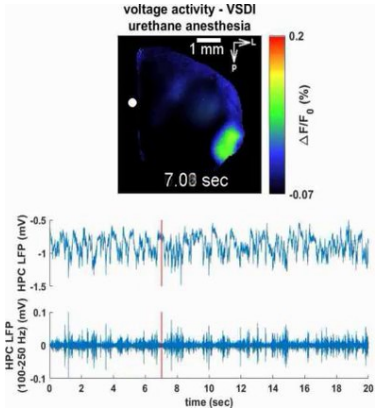
The ripple-timing study found a continuum of cortical activation times. Some cortical activity preceded the hippocampal event, while other activity followed it. This pattern differed from a strictly one-directional model in which ripples always arise independently in the hippocampus and then send information to a previously inactive cortex.

The temporal balance of hippocampal population activity was associated with the relative timing of cortical activation. Different events could therefore participate in different sequences of network communication. The observations were consistent with dialogue in which cortical activity sometimes contributes to the initiation of hippocampal reactivation rather than always serving as its passive destination.

Timing alone cannot identify every causal connection or determine the age of each memory involved. The study's direct result was the measured distribution of leading and lagging activity. This result constrains consolidation models by requiring them to accommodate several directions and timings of hippocampal–cortical coordination.



Karimi Abadchi et al. (2020), Fig. 6.



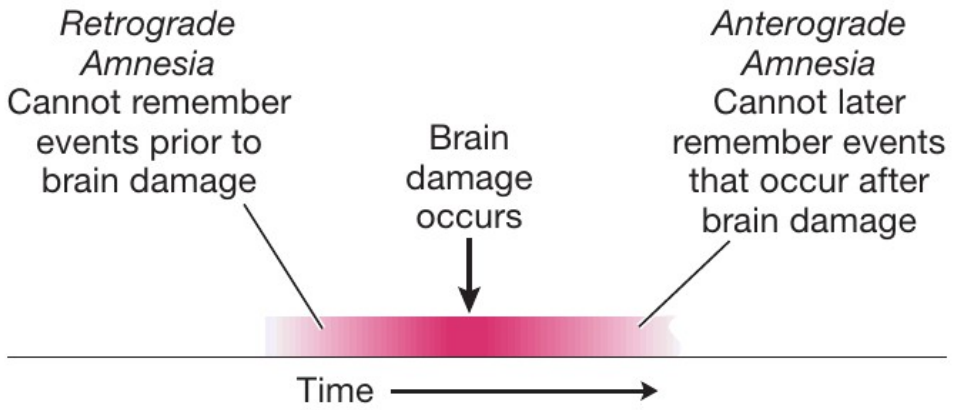
Karimi Abadchi et al. (2020), Video 1.

Amnesia can affect past and future learning differently

Anterograde amnesia is difficulty establishing lasting memories after the event causing the disorder. Retrograde amnesia affects information acquired before that event. A person can show both, and their extent can differ across memory types and time periods. The terms describe the relation to the injury or illness rather than a single severity scale.

Immediate retention can remain available when lasting new recollection is impaired. Familiar skills can also survive despite difficulty remembering recent experiences. Testing these capacities separately identifies the pattern of the disorder. A description such as memory loss is too broad to specify which information is unavailable and which forms of learning remain possible.

Medial-temporal and diencephalic disorders can produce different anatomical patterns of amnesia, and other conditions can impair retrieval through frontal or distributed network dysfunction. Clinical interpretation combines the history, neurological findings, and specific test results. The diagnosis cannot be inferred from one region's name without knowing the affected tissue and behavioral profile.



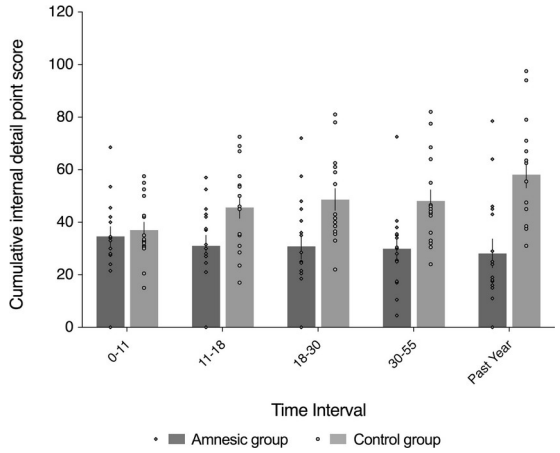
Carlson and Birkett (2022), Fig. 13.24, p. 440.

CA3 damage can reduce details of remote events

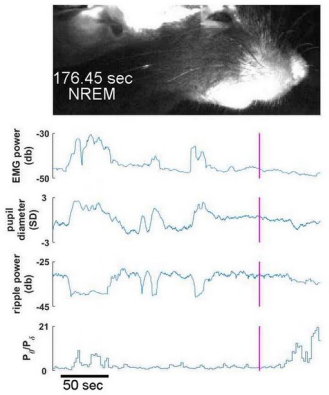
The human CA3 study assessed autobiographical memories from several periods of participants' lives. Interviewers separated internal episodic details from more general comments and knowledge. This allowed the researchers to examine the quality of a recollection rather than asking only whether the person could produce any statement about an old event.

Participants with CA3 damage produced fewer episodic details across several recent and remote intervals, although the earliest sampled childhood period was relatively preserved. Some affected memories concerned events long before the illness. Remote age therefore did not guarantee that richly contextual recollection was independent of hippocampal integrity.

The result qualifies a simple transfer account in which all old memories become fully detached from the hippocampus after a fixed interval. Different retained features can have different neural requirements. Testing detailed event reconstruction revealed a pattern that could be missed by scoring only a general fact or a broad sense of familiarity.



Miller et al. (2020), Fig. 4.



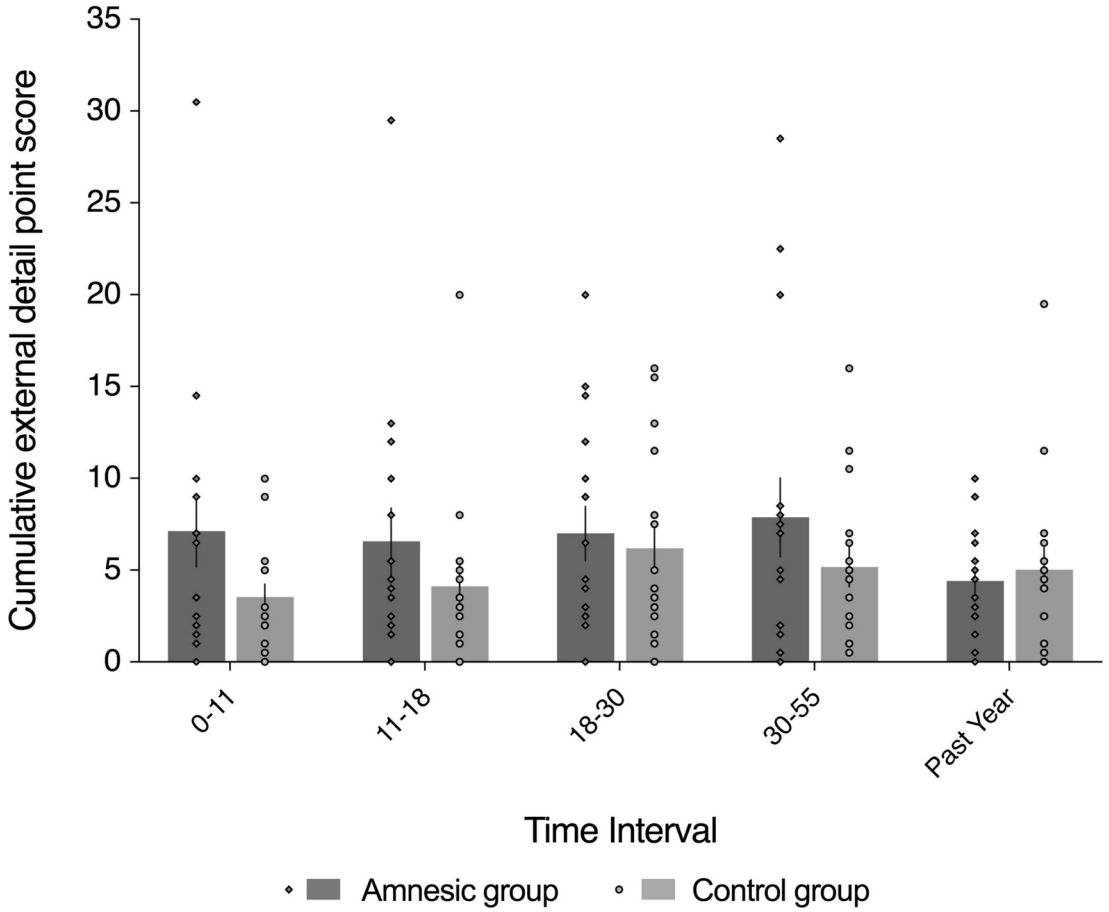
Karimi Abadchi et al. (2020), Video 2.

General knowledge can survive episodic-detail loss

The CA3 study also counted external details, including general personal-semantic information and material not specific to the chosen event. These details were relatively preserved compared with the loss of internal episodic content. Participants could therefore retain knowledge relevant to their lives while having difficulty reconstructing a particular experience in detail.

This distinction explains how someone can know that an event occurred without mentally revisiting it in the same way as before. The two responses draw on overlapping but distinguishable information. A memory assessment that asks only for the event's name or basic fact may underestimate a deficit in the contextual richness of recollection.

The investigators also examined functional integration within a medial-temporal network. Altered coordination was associated with autobiographical retrieval performance. This network measurement extended the anatomical finding by assessing how surviving regions communicated, while remaining distinct from the behavioral measurement of what details a participant could actually report.



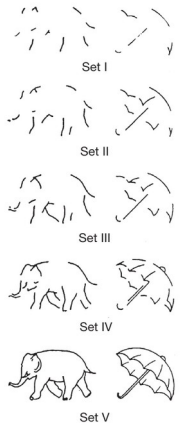
Miller et al. (2020), Fig. 5.

Practice can improve skills without recollection

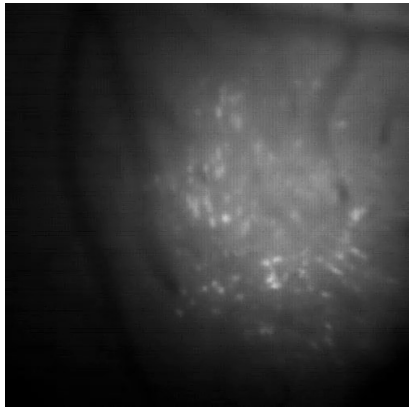
Patients with severe anterograde amnesia can improve on selected motor, perceptual, and stimulus–response tasks. Improvement may occur even when they do not remember having practiced. This preserved learning shows that the neural changes supporting a skill or familiarity effect can remain available when the formation of a detailed new event memory is impaired.

A serial reaction-time task can measure changes produced by repeated sequences, while mirror tracing measures improvement in a visuomotor skill. Broken-picture recognition can reveal perceptual learning. Each task has additional demands, so the degree of preservation depends on the task and the person’s particular lesion rather than on a universal rule that all nondeclarative learning is intact.

Conscious report and performance therefore provide complementary evidence. A participant may improve without an accompanying recollection, or remember a training episode without performing the skill well. Comparing the two identifies whether the experiment concerns knowledge of the experience, a change in behavior produced by it, or both.



Carlson and Birkett (2022), Fig. 13.27, p. 443.



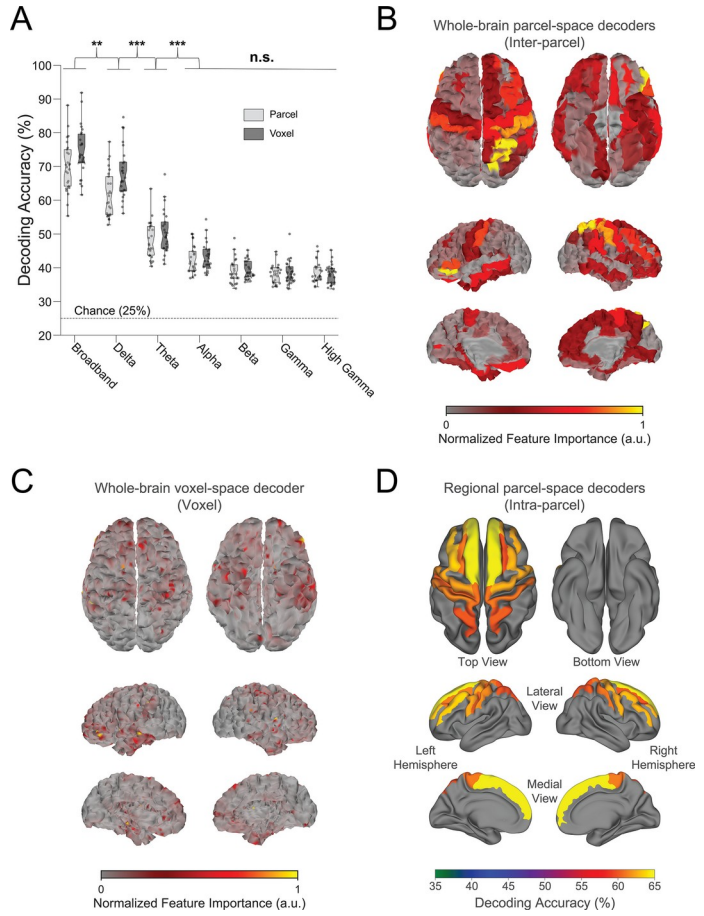
Li et al. (2024), Video 5.

Human skill can be measured in neural movement patterns

A human study recorded magnetoencephalographic activity while participants practiced a sequence of finger movements. Magnetoencephalography measures magnetic fields associated with synchronized neural electrical activity. Decoding analyses tested whether the recorded patterns could distinguish finger identities, providing a way to follow how action representations changed during early learning.

The same finger movement could occur at different positions within a sequence. This allowed the investigators to ask whether its neural representation remained identical or became sensitive to sequence context. Movement identity and contextual position were therefore separated rather than assuming that a finger always had one fixed representation during every skill.

The analysis identified contributions from different frequencies and cortical regions to movement decoding. These measurements characterized a changing neural representation, while behavioral speed and accuracy measured the acquired skill. A decoder's success was interpreted as information available in the recorded signal, not as proof that the brain used exactly the same algorithm.



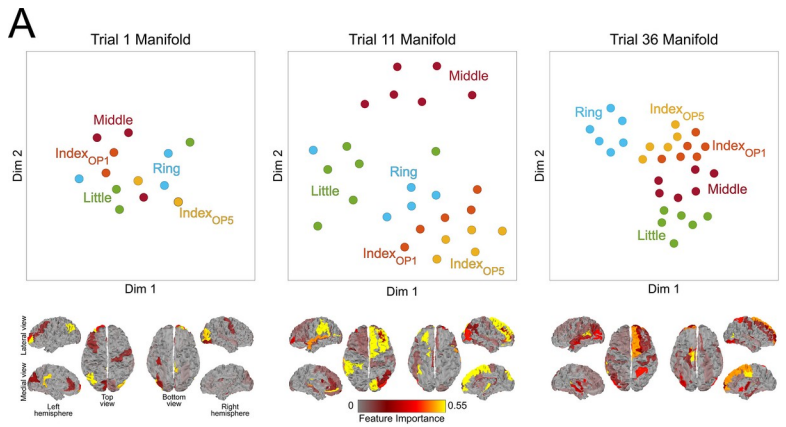
Dash et al. (2025), Fig. 2.

Sequence context becomes clearer during learning pauses

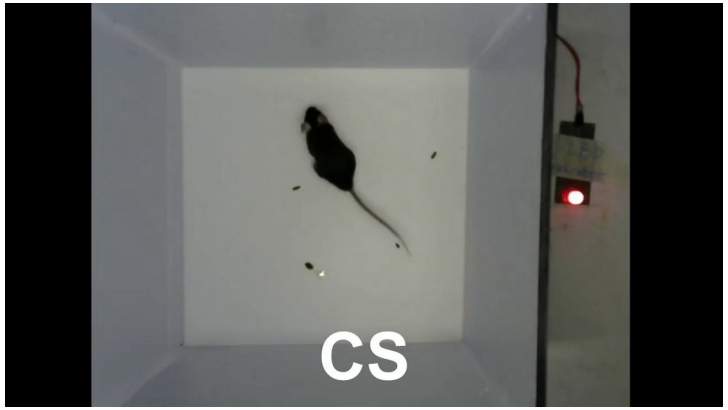
The human sequence-learning study found that representations of the same action in different sequence positions progressively differentiated during early learning. The change was especially evident across rest intervals between practice periods. Greater contextual differentiation during those pauses was associated with larger skill gains.

Changes during active practice did not show the same relationship to learning. This distinguished simply producing repeated movements from the offline evolution of their representation. The neural regions contributing to the change also shifted as performance approached a plateau, indicating that early improvement and stable skilled performance involved different population patterns.

The contextual distinction persisted into the next day and showed specificity for the practiced sequence. A neural change observed across a brief pause could therefore participate in a more lasting skill representation. The result concerned the organization of action information over time, rather than a new ability to consciously recount every movement of the practice session.



Dash et al. (2025), Fig. 4.



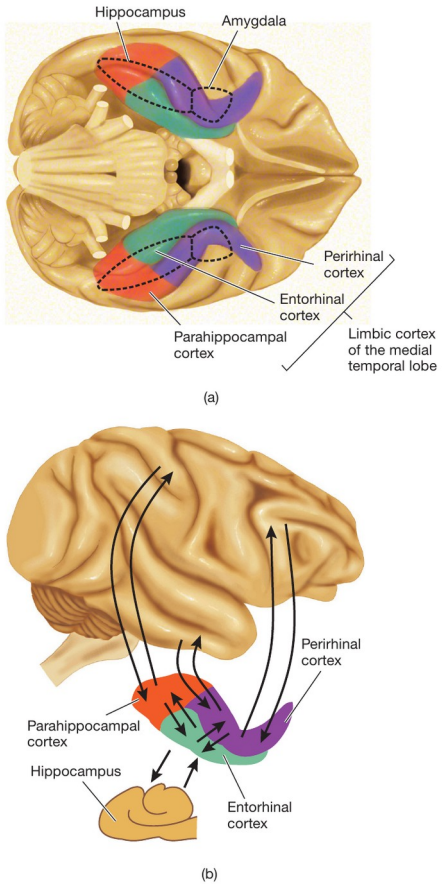
Feng et al. (2021), Video 10.

Retrieval depends on cues and can alter later remembering

Retrieval uses retained information under the conditions of a current test or experience. A cue can reactivate some features of an event while leaving others unavailable. Interference occurs when competing information affects which response is selected, so unsuccessful recall can result from competition rather than complete loss of every stored trace.

Retrieving information can also change later performance. Repeated recall can strengthen access, promote integration with new information, or alter competition among related memories. These effects depend on the procedure and retained content. They should be measured rather than assumed to follow from every act of remembering in the same way.

Reconsolidation refers to processes through which a reactivated memory can become susceptible to change and subsequent restabilization under particular conditions. This is distinct from merely presenting a reminder or adding a new association. Establishing the process requires appropriate timing and comparisons that distinguish modification of an old memory from separate new learning.



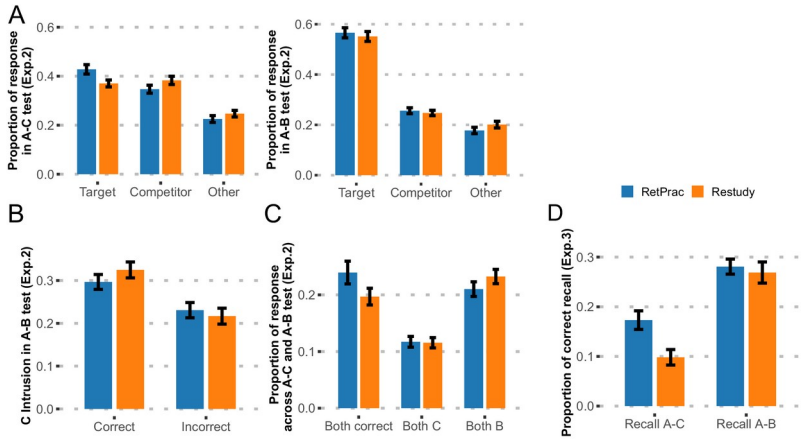
Carlson and Birkett (2022), Fig. 13.20, p. 434.

Retrieval practice can improve memory updating

A human study taught participants an initial set of paired associations and then introduced new associations sharing an earlier cue. The new pairing could compete with the old one during the final test. The investigators compared retrieving the new information with restudying it, testing how the learning procedure changed later updating and intrusion.

Retrieval practice strengthened the new target associations and reduced intrusions from the older competitors. Separate tests showed that this was not simply explained by wholesale suppression of the original memories. The result distinguished improving selection of the updated response from erasing all information associated with the earlier pairing.

The researchers measured both target recall and competitor-related errors across conditions. This allowed a useful update to be identified by its actual behavioral consequence. Remembering the new association more reliably and retaining the old association somewhere in memory were not mutually exclusive outcomes.



Ye et al. (2020), Fig. 2.



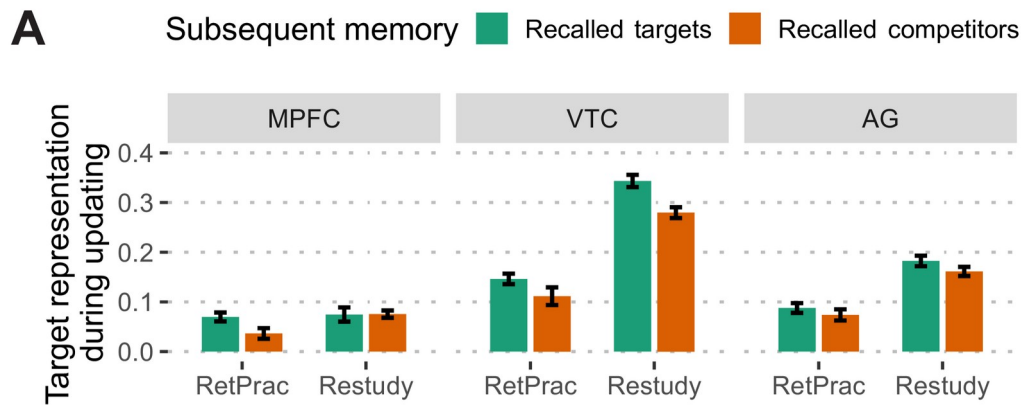
Feng et al. (2021), Video 11.

Updating changes prefrontal memory representations

The memory-updating study used functional imaging and pattern analysis to examine target and competitor information in medial prefrontal cortex. Patterns during updating were compared with the later memory test. This measured how information represented during the procedure related to subsequent retrieval rather than inferring the entire mechanism from final accuracy alone.

Under retrieval practice, stronger target representation during updating predicted later memory, and the final test showed enhanced target-related information. Competitor evidence could remain present as well. Successful updating therefore involved changes in how related information was organized and used, not necessarily the complete removal of the competing representation.

Pattern evidence is an analysis of information in a population-level imaging signal. It does not identify a single neuron containing the target word or establish that the old memory was physically destroyed. The observed coexistence of target and competitor information helped explain why reduced behavioral intrusion could occur without equivalent suppression of the old association.



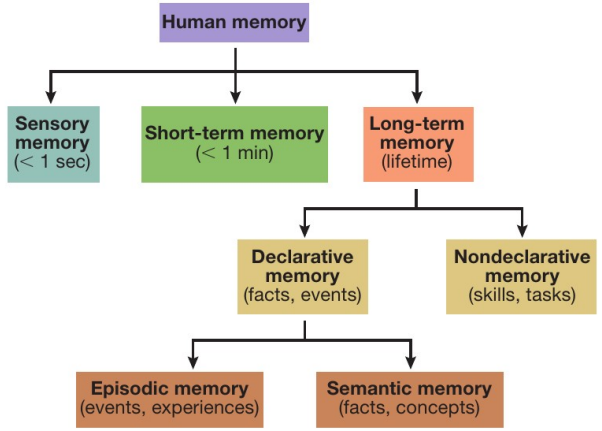
Ye et al. (2020), Fig. 4.

Forgetting can reflect storage change or failed access

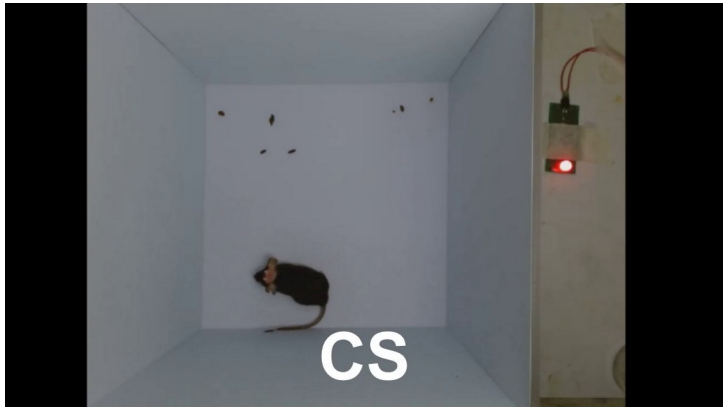
Forgetting is a decline in the influence of previously acquired information on behavior or report. It can arise through weakened storage, altered retrieval conditions, interference, or changes in how a trace is expressed. A failed test therefore establishes that information was not successfully used under those conditions, not that every underlying modification has been erased.

Engram experiments identify neural populations recruited during learning and test whether later reactivation contributes to recall. An engram is a biological trace associated with memory, not a literal miniature recording of an experience. Activity-dependent labeling, selective inhibition, and selective activation provide different ways to examine whether a recruited population remains capable of influencing behavior.

A reminder can sometimes restore performance after apparent forgetting. Such recovery is evidence about residual information and access, but its interpretation depends on what the reminder supplies. The experiment must distinguish reactivating an earlier trace from teaching the entire task again during the reminder itself.



Carlson and Birkett (2022), Fig. 13.9, p. 422.



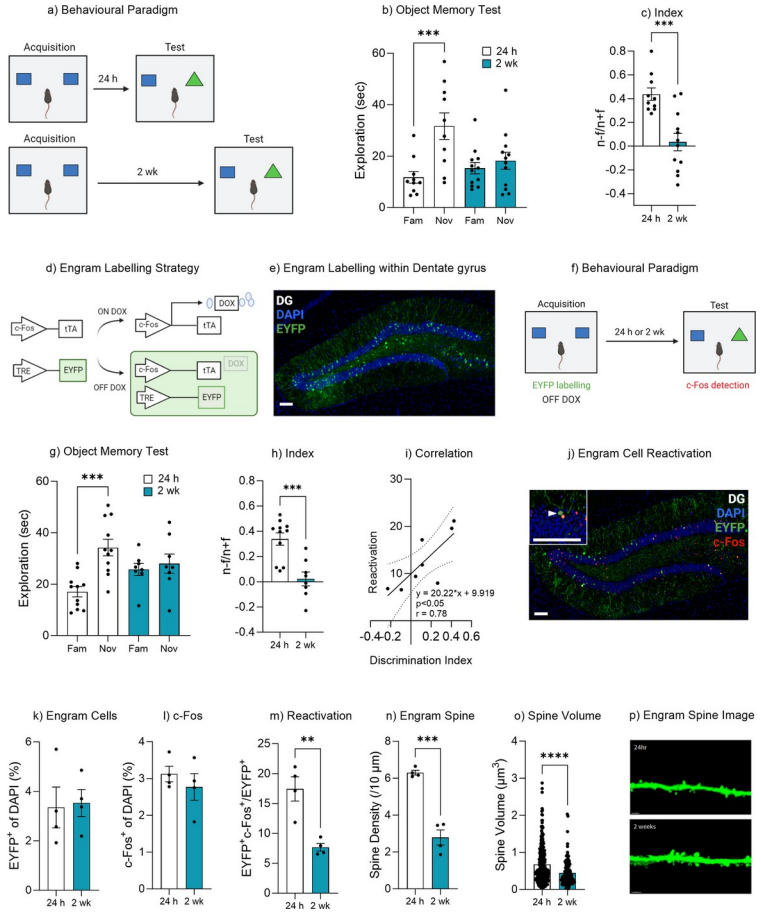
Feng et al. (2021), Video 12.

Object memory can become unavailable over time

A mouse study tested memory for objects after different retention intervals. Mice naturally explore novelty, allowing preference for a new object to indicate use of information about an earlier familiar one. The study combined this behavioral measure with labeling of a population recruited during the learning episode.

At a longer delay, the behavioral expression of object memory declined. The researchers then manipulated engram activity and other conditions that could influence retention or recall. Direct stimulation could facilitate expression, while inhibition could prevent it, showing that the activity of the identified population contributed to performance under the tested conditions.

The object task required exploration and comparison, so behavior was assessed alongside the experimental conditions controlling access to the trace. The central distinction was between a memory that failed to guide spontaneous choice and a recruited population that could still influence that choice when its activity was changed.



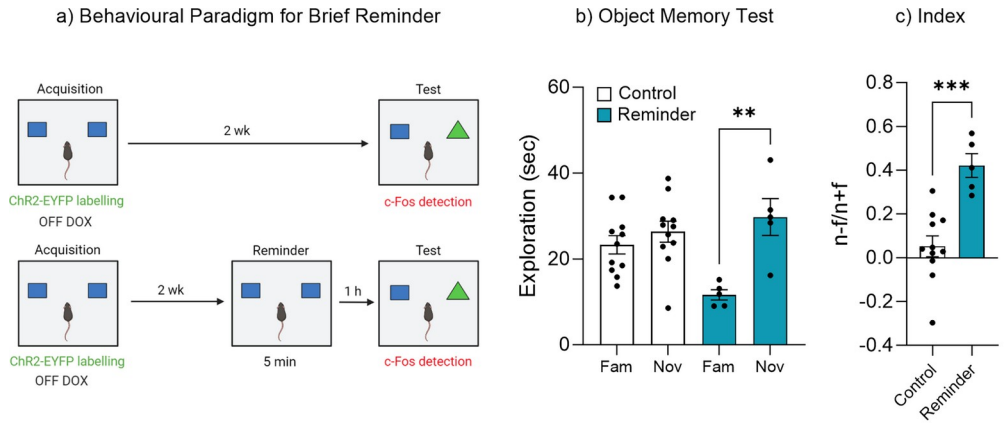
O'Leary et al. (2024), Fig. 1.

A brief reminder can restore an earlier object memory

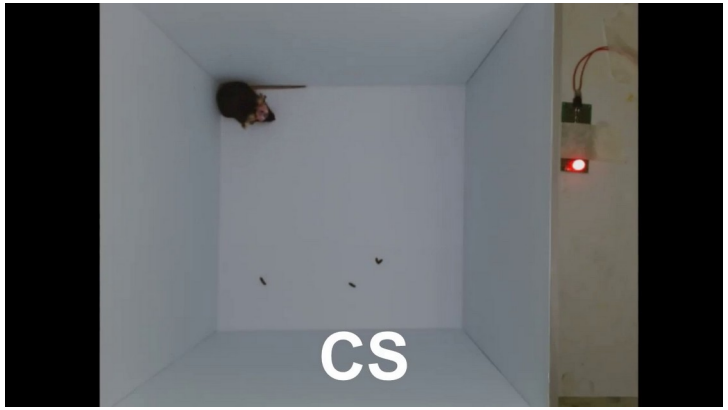
The object-memory study provided a brief exposure to the original stimuli after the ordinary test interval would produce apparent forgetting. This reminder improved later expression of the familiar-object memory. The experiment examined whether information could become accessible again without requiring a full repetition of the original learning schedule.

Reactivation of the labeled population was evaluated together with the recovered behavioral response. The findings were consistent with a trace shifting between more accessible and less accessible states. This interpretation concerned the tested object memory and the particular interventions, rather than establishing that every forgotten human experience remains perfectly stored and recoverable.

The study also used conditions that accelerated or prevented forgetting. Different experiences and physiological manipulations changed the probability that the trace influenced later behavior. Retention was therefore a dynamic property of the organism and its environment, not simply an unchanging label assigned at the end of the first exposure.



O'Leary et al. (2024), Fig. 4.



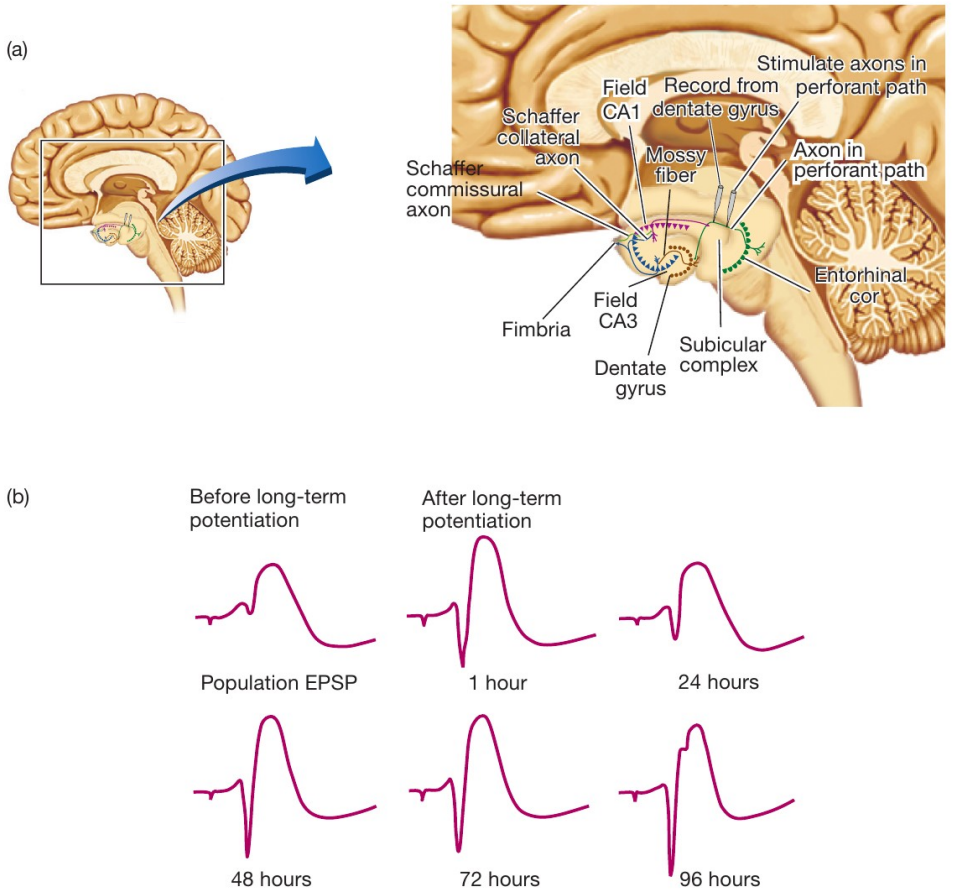
Feng et al. (2021), Video 13.

Long-term potentiation strengthens tested synaptic input

Long-term potentiation, or LTP, is a lasting increase in a synaptic response after an induction procedure. In a hippocampal preparation, a weak test pulse measures the baseline response of a pathway. A stronger patterned stimulation is then delivered, and the same test pulse is repeated to determine whether transmission has changed.

In many commonly studied pathways, induction depends on coincident synaptic activity and postsynaptic depolarization. Rapid inputs can summate, producing a larger voltage change than the same inputs spaced farther apart. The resulting calcium signal activates processes that increase the effectiveness of the active connections.

LTP is a cellular phenomenon used to investigate mechanisms relevant to learning. A larger field response does not by itself specify what an intact animal remembers. Behavioral, circuit, and cellular experiments are needed to connect a particular synaptic change to the encoding, persistence, or expression of a defined memory.



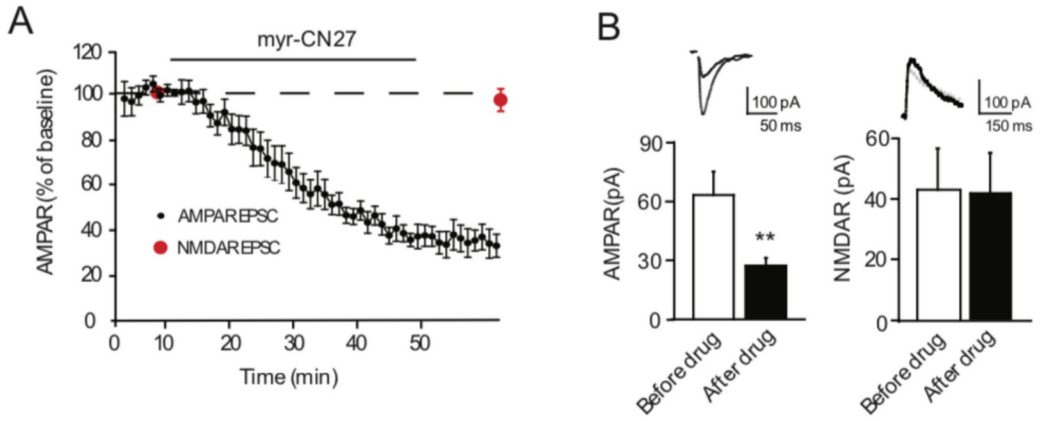
Carlson and Birkett (2022), Fig. 13.30, p. 448.

CaMKII contributes to maintaining synaptic strength

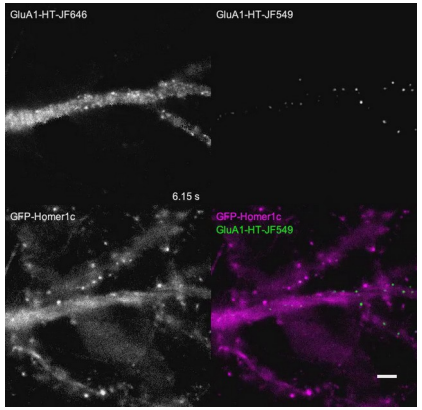
Calcium–calmodulin-dependent protein kinase II, or CaMKII, is an enzyme activated by a calcium-binding regulatory protein. A kinase transfers phosphate groups to other proteins, changing their properties or interactions. CaMKII is abundant at excitatory synapses and has been studied both during the induction of potentiation and after synaptic strength has increased.

A study applied a CaMKII inhibitor while recording AMPA-receptor-mediated synaptic transmission. The effect depended on CaMKII-related conditions and reduced the measured response. This tested an ongoing contribution to established synaptic function, rather than only asking whether the enzyme was activated briefly during the original induction procedure.

The investigators distinguished different receptor-mediated currents and used genetic and pharmacological comparisons. These controls helped identify which component of transmission changed after inhibition. The direct outcome concerned the strength of recorded synaptic responses; claims about a person’s stored experiences require evidence beyond this cellular preparation.



Tao et al. (2021), Fig. 1.



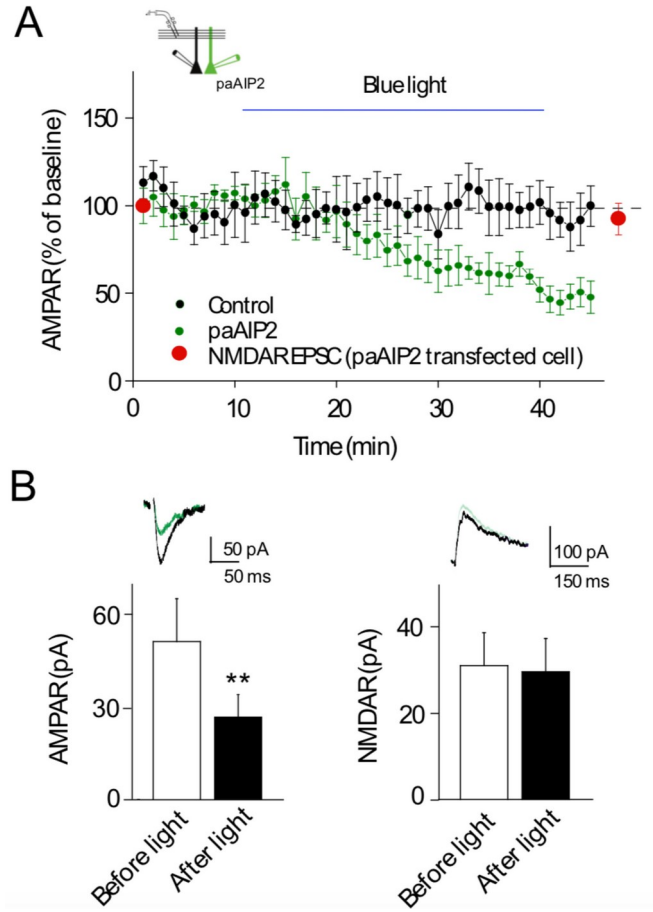
Wong et al. (2024), Video 1.

Light-controlled inhibition tests an ongoing kinase role

The CaMKII study also used a light-activated inhibitor to change kinase function with temporal control. The approach allowed inhibition to be applied after a synaptic state had already formed. Comparing the response before, during, and after the intervention tested whether ongoing enzyme-related function helped maintain the established transmission.

Light activation of the inhibitor reduced AMPA-mediated synaptic responses through a CaMKII-dependent mechanism. The study also examined how transient inhibition affected later potentiation. The timing separated a contribution to maintaining the existing state from a contribution limited to the first moments when stimulation initiated the change.

These experiments inform a molecular account of synaptic persistence, but the target operates within a larger postsynaptic system. Receptor anchoring, protein interactions, and other signaling pathways also determine the measured response. A maintenance requirement in a tested pathway does not make the enzyme an independently readable record of every feature of a memory.



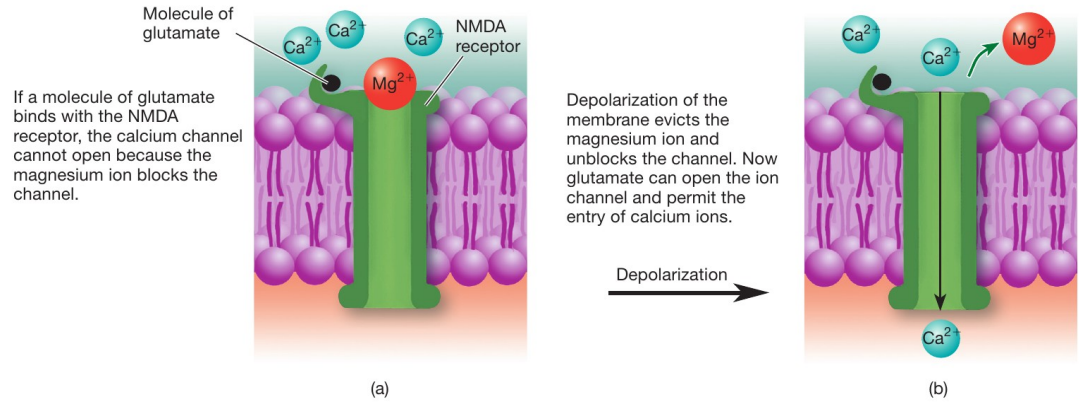
Tao et al. (2021), Fig. 3.

NMDA receptors detect transmitter during depolarization

NMDA receptors are glutamate-gated ion channels whose current also depends on membrane voltage. Near resting potential, magnesium blocks much of the channel. Depolarization reduces this block, so glutamate binding during sufficient postsynaptic activation can permit calcium to enter. The receptor therefore responds to a combination of chemical input and electrical state.

AMPA receptors can provide the initial fast depolarization that helps relieve the magnesium block. Several active inputs can summate, and back-propagating or local dendritic events can add depolarization. The relevant coincidence occurs at the receiving cell, allowing activity at selected synapses to produce a calcium signal under conditions different from weak isolated input.

Calcium entry then recruits intracellular processes rather than directly becoming a stored fact or image. Its concentration, location, and duration influence which enzymes respond. Blocking NMDA receptors can prevent induction in many preparations, while the maintenance and expression of an already established change can have different requirements.



Carlson and Birkett (2022), Fig. 13.33, p. 449.



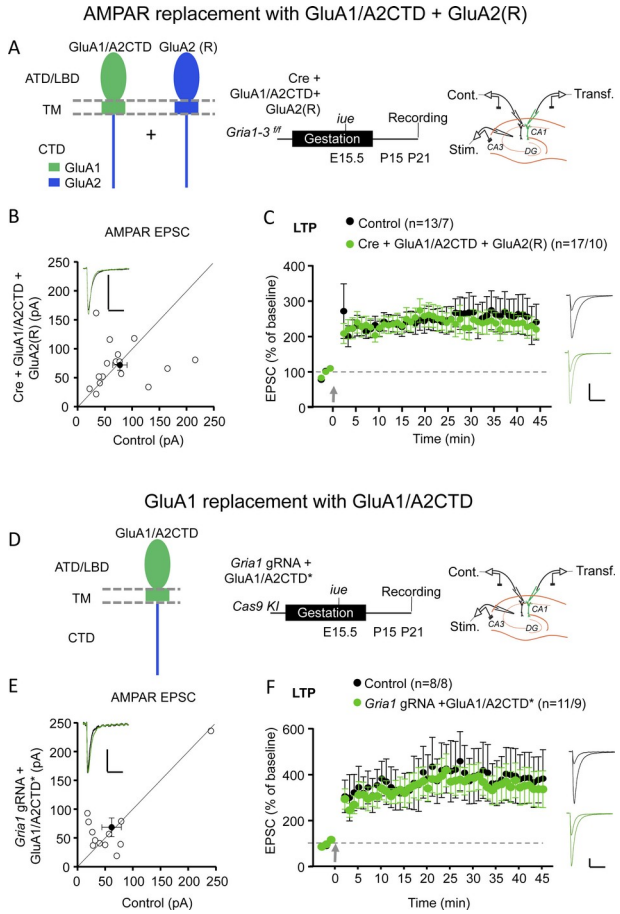
Wong et al. (2024), Video 2.

An AMPA receptor tail is not required for every LTP protocol

An experimental study tested the proposed necessity of the intracellular tail of the GluA1 AMPA-receptor subunit. The tail had been considered an important site for interactions controlling receptor delivery during potentiation. The researchers altered this part of the receptor and measured whether the synapse could still express long-term potentiation.

Receptors with the tested tail modification still supported LTP under the experimental conditions. The result separated the requirement for functional AMPA receptors from the stronger claim that one particular tail sequence was indispensable for potentiation. Receptor recruitment could occur through mechanisms not eliminated by that modification.

The study favored an account in which potentiation creates or changes synaptic binding opportunities that capture mobile receptors. Such an account still requires receptor movement and retention, but it does not assign all of those operations to one mandatory segment of GluA1. Testing a specific molecular prediction refined the mechanism rather than rejecting synaptic plasticity itself.



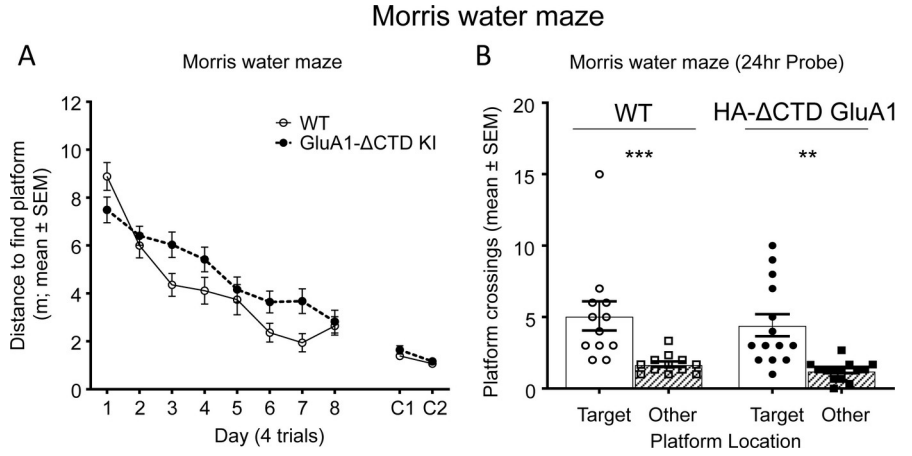
Díaz-Alonso et al. (2020), Fig. 1.

Spatial learning can survive the tested GluA1-tail change

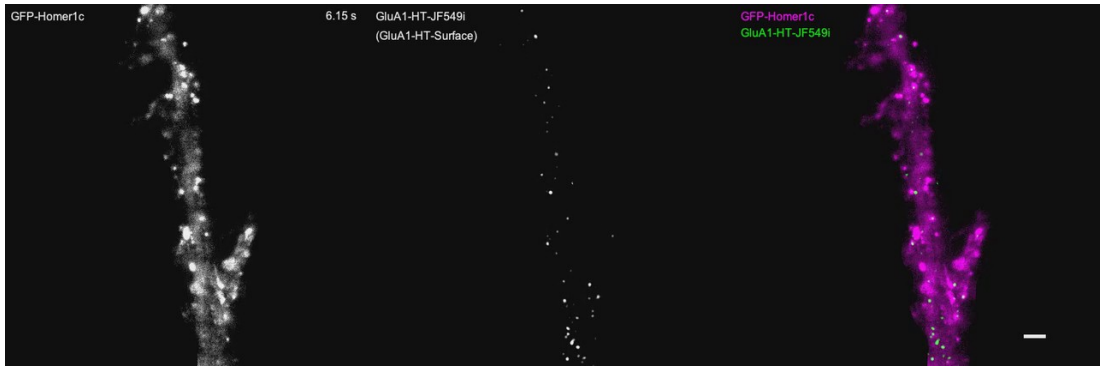
The AMPA-tail study also examined spatial learning in mice with the relevant receptor modification. In the Morris water maze, animals learn to locate a hidden platform using information from the surrounding environment. Training performance and later search behavior provided behavioral measures separate from the electrical recordings used to assess LTP.

The tested animals showed preserved spatial learning and memory in this assay. This behavioral outcome accompanied the preserved potentiation measured in the study. The modification therefore did not abolish every route by which AMPA receptors could contribute to synaptic strengthening or the tested form of spatial learning.

A preserved task does not show that the altered molecular region has no function under any condition. It shows that the proposed absolute requirement did not hold in these preparations and tests. The experiment distinguished a specific molecular hypothesis from the broader evidence that changing excitatory synapses contributes to learning.



Díaz-Alonso et al. (2020), Fig. 3.



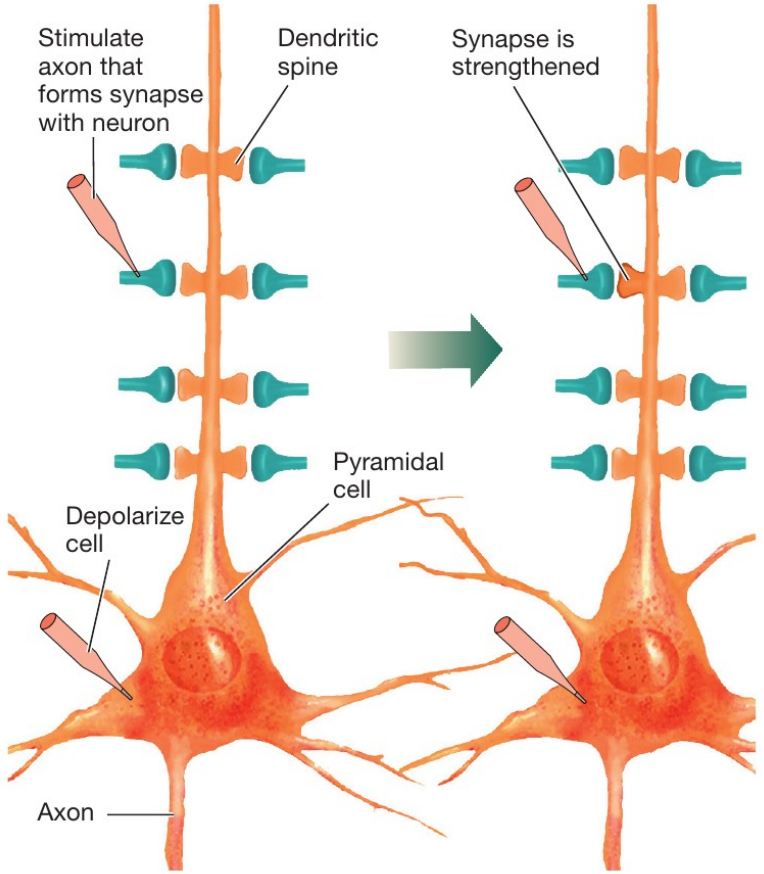
Wong et al. (2024), Video 3.

Calcium activates enzymes in the receiving synapse

When calcium enters an active dendritic spine, it can bind calmodulin and activate enzymes such as CaMKII. These enzymes modify proteins involved in receptor function, anchoring, and the organization of the postsynaptic region. The result can be a stronger response to the same later amount of glutamate released from the presynaptic terminal.

Calcium also influences the balance of kinase and phosphatase activity. A kinase adds phosphate groups, while a phosphatase removes them. Different patterns of calcium can therefore favor different changes rather than producing a fixed response whenever any calcium appears. Location and duration determine which nearby proteins are reached and how the signaling evolves.

The spine is connected to a dendritic shaft containing transport machinery and internal membranes. Strengthening can involve delivery of additional receptors as well as changes to receptors already present. Explaining plasticity therefore requires a sequence from the coincidence signal to protein modification, receptor availability, and the later electrical response.



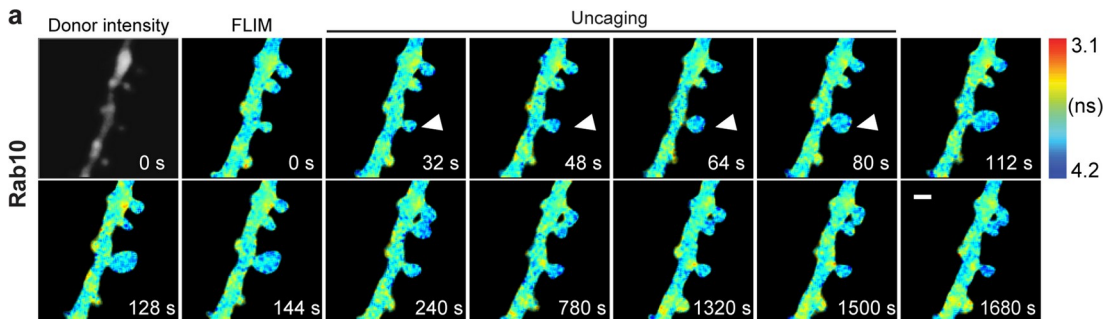
Carlson and Birkett (2022), Fig. 13.32, p. 449.

Rab10 activity falls in a spine undergoing enlargement

A recent study measured Rab-protein signaling during structural potentiation of individual dendritic spines in hippocampal tissue. Rab proteins regulate membrane trafficking by cycling between activity states. Fluorescence-lifetime sensors allowed the investigators to follow these changes near a stimulated spine rather than measuring only a whole-cell average after the experiment.

Rab10 became persistently less active in the stimulated spine, while Rab4 showed a more transient increase. The signals therefore changed in opposite directions and over different intervals. This timing helped identify how distinct trafficking regulators could contribute to the immediate enlargement and the longer-lasting development of the structural response.

The experiment paired a controlled local induction procedure with imaging of spine volume and molecular activity. The recorded change in Rab signaling was localized in relation to the stimulated structure. It supplied a candidate intracellular step between synaptic activation and altered delivery of membrane components, rather than treating the enlarged spine as an unexplained final shape.



Wang et al. (2025), Fig. 2.



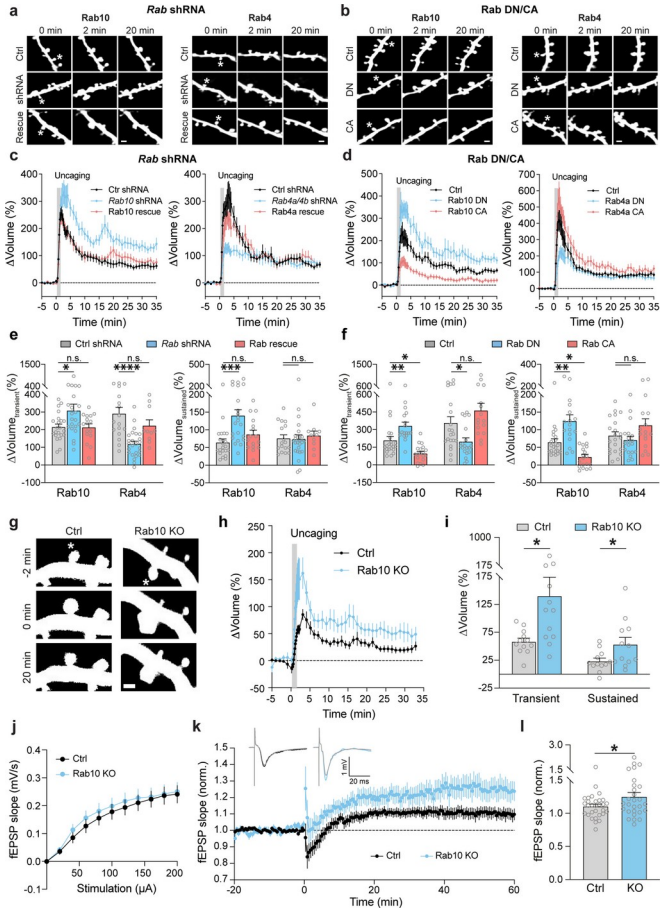
Wong et al. (2024), Video 4.

Opposing trafficking regulators shape spine plasticity

The Rab study inhibited or deleted Rab10 and also disrupted Rab4 signaling. These interventions tested the direction of their contribution rather than relying only on correlations during stimulation. The investigators measured structural potentiation, electrical potentiation, and AMPA-receptor trafficking as related but distinguishable consequences.

Reducing Rab10 enhanced spine enlargement and relevant measures of potentiation, while disrupting Rab4 weakened the early structural phase and reduced receptor trafficking. The regulators therefore did not provide interchangeable transport functions. Their opposite effects helped explain why one activity signal fell as another briefly rose during the same induction period.

AMPA receptors must reach and remain at appropriate membrane sites to increase transmission. Rab-dependent trafficking changes the supply route, while other proteins regulate capture and anchoring. A local spine can therefore become more responsive through coordinated changes in several transport steps, with different regulators controlling different phases of the response.



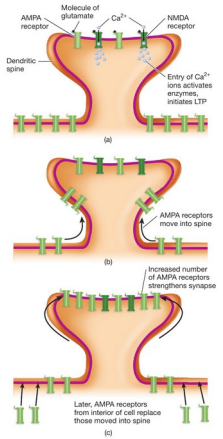
Wang et al. (2025), Fig. 4.

AMPA delivery increases excitatory responses

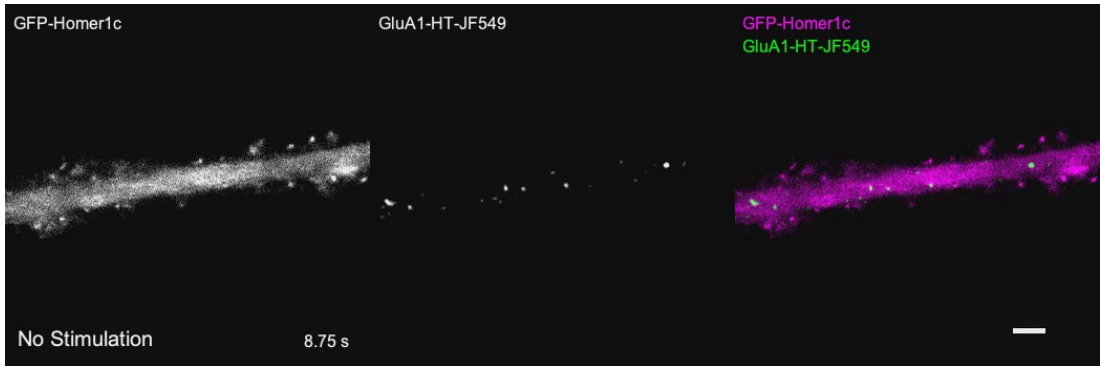
AMPA receptors open rapidly when glutamate binds and carry an excitatory current at many central synapses. Increasing the number or effectiveness of these receptors can increase the postsynaptic response to later input. Receptors move within the membrane and can also be delivered from internal vesicles, so their local availability is actively regulated.

Exocytosis occurs when a vesicle fuses with the cell membrane and adds its membrane proteins to the surface. Endocytosis removes membrane components into internal compartments. The balance between insertion, removal, lateral movement, and anchoring determines how many functional receptors remain near a synapse after an induction event.

Receptor delivery requires transport through the dendrite as well as events inside a spine. A change in synaptic activity can influence nearby internal stores and the cytoskeleton. The receiving synapse is therefore maintained by ongoing molecular traffic rather than by receptors becoming permanently motionless after the first learning-related stimulation.



Carlson and Birkett (2022), Fig. 13.35, p. 451.



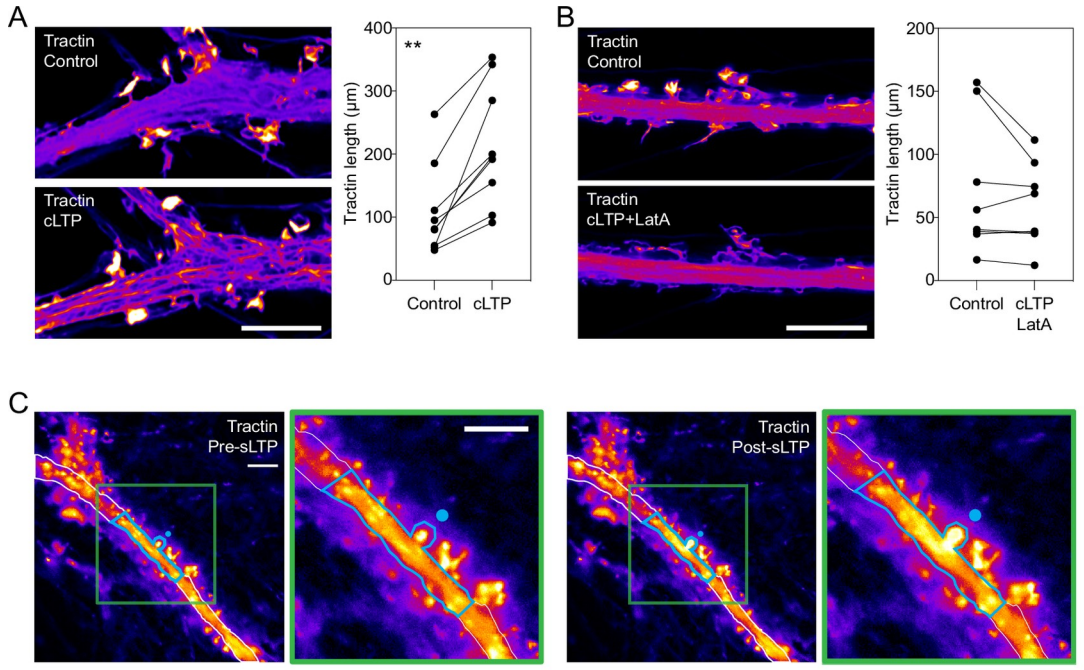
Wong et al. (2024), Video 5.

Actin can retain receptor vesicles near active sites

A study labeled endogenous GluA1 AMPA-receptor subunits in cultured rat hippocampal neurons. Endogenous labeling followed receptors produced from their normal genetic location rather than relying only on large amounts of an added receptor construct. Individual receptor-containing vesicles were tracked as synaptic activity induced local structural plasticity.

After stimulation, vesicles became more confined near the active region of the dendrite. Increased polymerization of actin, a structural protein that forms filaments, contributed to this restriction of movement. The local cytoplasm therefore changed the route available to receptor-containing cargo rather than simply increasing the speed of all transport throughout the cell.

Experiments reducing actin polymerization diminished the relevant confinement. This manipulation connected the observed vesicle behavior with a specific physical component of the cell. The mechanism could concentrate a receptor reservoir near active synapses, increasing local availability without requiring every vesicle to be transported directly from the cell body at the moment of stimulation.



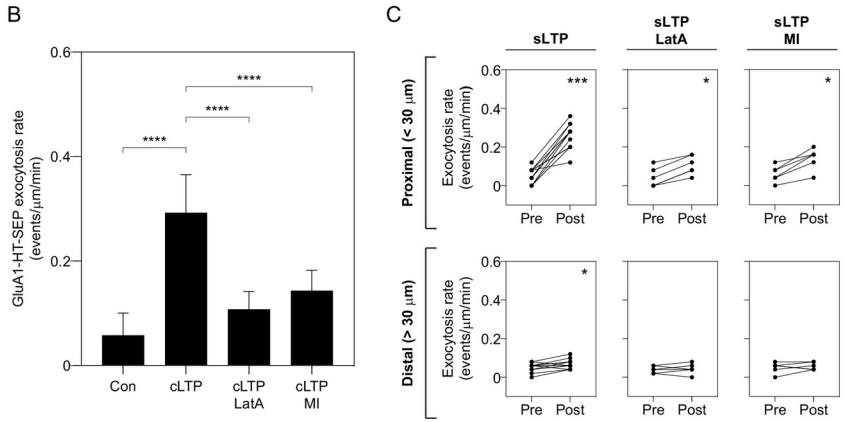
Wong et al. (2024), Fig. 3.

Actin and myosin help receptors reach exocytic sites

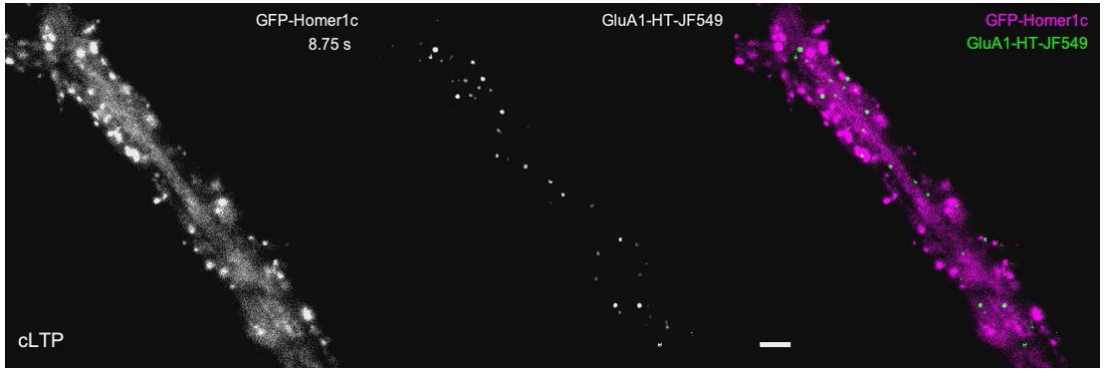
The GluA1-trafficking study used a pH-sensitive fluorescent tag to detect receptor exposure at the cell surface during vesicle fusion. A signal associated with exocytosis could therefore be distinguished from a vesicle merely moving inside the dendrite. This allowed delivery to be measured separately from intracellular transport and confinement.

Synaptic stimulation increased receptor exocytosis, and the response depended on actin polymerization and myosin activity. Myosin is a motor protein that uses chemical energy to move along actin filaments. The same local filament network that restricted long-range vesicle motion could therefore assist movement toward sites where cargo was inserted into the membrane.

The mechanism involved successive physical operations: actin altered the local transport environment, receptor-containing vesicles accumulated nearby, and motor-dependent movement supported surface delivery. These observations were made in the cultured-neuron preparation. They identify a cellular process relevant to strengthening without claiming that the microscope directly displayed a particular learned experience.



Wong et al. (2024), Fig. 5.



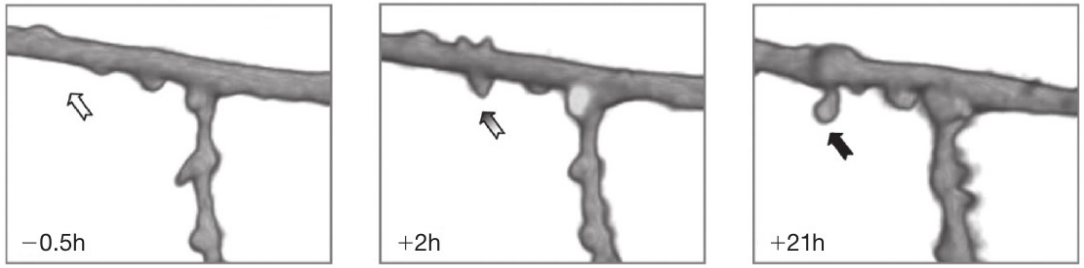
Wong et al. (2024), Video 6.

Spines change structure while synapses change function

Dendritic spines are small protrusions that receive many excitatory synapses. Their shapes and sizes vary, and activity can influence their growth, shrinkage, or persistence. Structural change can accompany altered receptor content and synaptic strength, but spine size and electrical transmission are different measurements rather than identical definitions of memory.

New spines can emerge and existing spines can be removed as circuits change through experience. Some persist while others are transient. Longitudinal imaging is needed to follow the same structure across time, because a single image cannot reveal whether a spine recently appeared, remained stable, or was about to disappear.

Other cells interact with these structures. Astrocytes influence the synaptic environment, and microglia move processes through the surrounding tissue. Studying these contacts extends the mechanism beyond the two neurons forming a synapse, while requiring experiments to distinguish a correlated contact from a necessary cause of structural change.



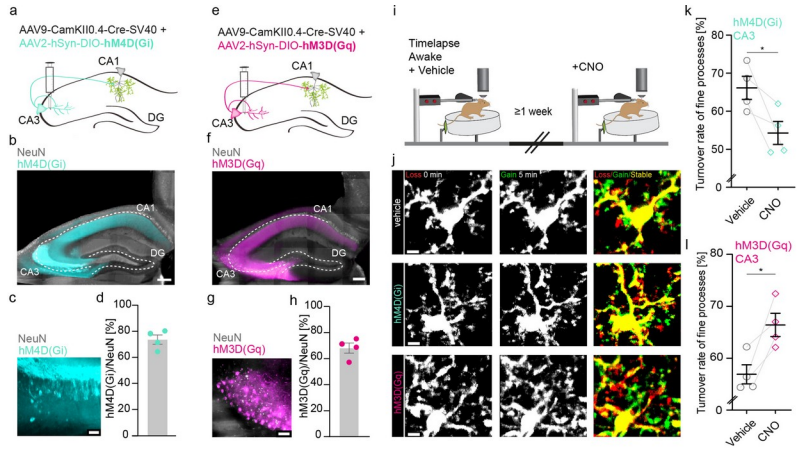
Carlson and Birkett (2022), Fig. 13.36, p. 452.

Hippocampal input changes microglial movement

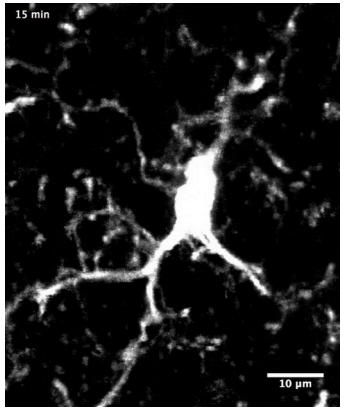
A mouse study repeatedly imaged microglia and dendritic spines in hippocampal CA1. Microglia are resident immune-related cells whose processes extend and retract through neural tissue. The investigators compared waking and anesthetized conditions and manipulated CA3 input to CA1 to test whether neuronal activity altered this local surveillance.

Increasing or decreasing CA3 input changed microglial process motility in opposite directions. This showed that the movement of a non-neuronal cell population could respond to activity in a defined hippocampal pathway. The experiment did not require assuming that microglia recognized the semantic content of a memory; it measured their physical response to circuit activity.

The comparison between awake and anesthetized imaging also changed observed spine dynamics. Physiological state therefore influenced the cellular events visible in the preparation. Keeping that state explicit helps distinguish ordinary waking plasticity from changes measured while neural activity and behavior have been altered by anesthesia.



Nebeling et al. (2023), Fig. 2.



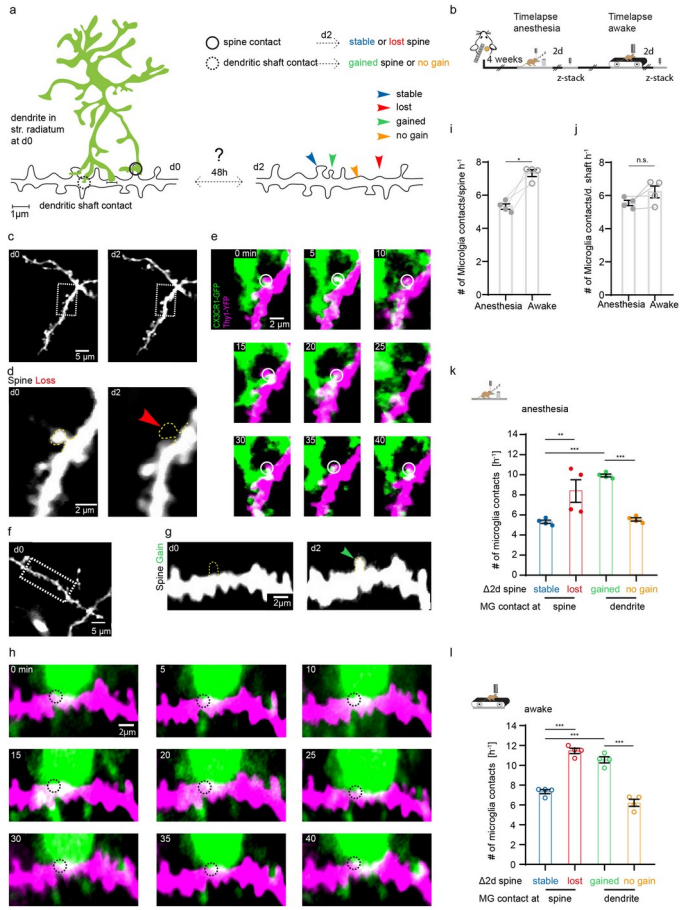
Nebeling et al. (2023), Video 1.

Microglial contacts are associated with spine fate

The hippocampal-imaging study followed contacts between microglial processes, dendrites, and individual spines. Repeated observations identified which structures persisted, disappeared, or emerged. This temporal information allowed contact history to be compared with subsequent structural fate rather than assuming that every nearby microglial process had the same effect.

Contact frequency was associated with differences in spine stability and remodeling. The result described a relationship measured across the observed structures. It did not establish that each contact directly caused elimination or growth, because activity and other local conditions could influence both microglial behavior and spine change.

The study therefore adds a specific cellular interaction to the biology of circuit remodeling: neuronal activity alters microglial motility, and contact patterns accompany different structural outcomes. Demonstrating an additional requirement for a particular learned behavior would need an intervention linking that interaction to the memory task itself.



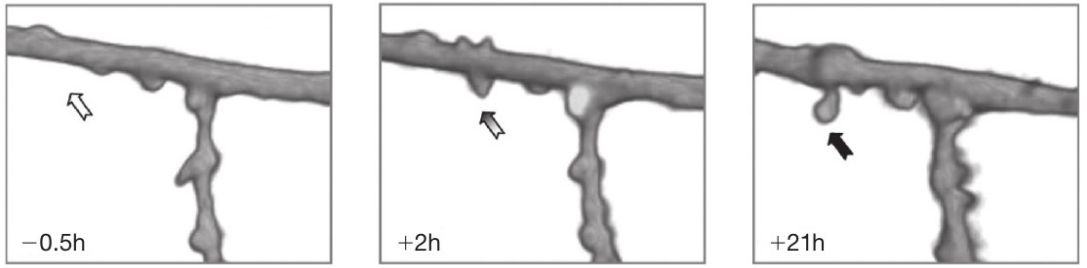
Nebeling et al. (2023), Fig. 4.

Persistent plasticity recruits new protein production

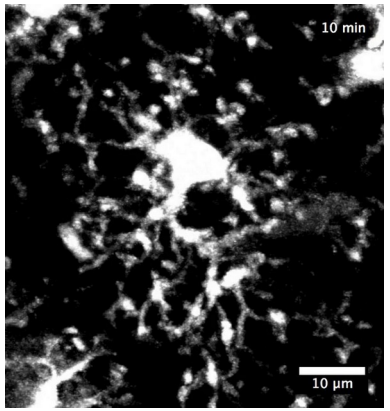
Some forms of potentiation last after the short-lived signaling produced during induction has declined. Protein synthesis can contribute to these later phases by supplying components needed to maintain altered synaptic organization. Experiments distinguish early and late phases by their time course and their different responses to inhibitors of translation or gene expression.

A protein can have a brief molecular lifetime while the altered cellular state persists through replacement and interacting feedback. Long-lasting synaptic change therefore does not require the same individual molecule to remain intact for decades. Receptor turnover, structural maintenance, and regulated protein production can preserve a modified system as its components change.

Candidate maintenance mechanisms must be tested under both acute and developmental manipulations. Removing a protein throughout life can allow compensating pathways to develop, while inhibiting it after learning tests the established system more directly. Different outcomes can reveal adaptation rather than proving that one experiment simply invalidates every other role of the molecule.



Carlson and Birkett (2022), Fig. 13.36, p. 452.



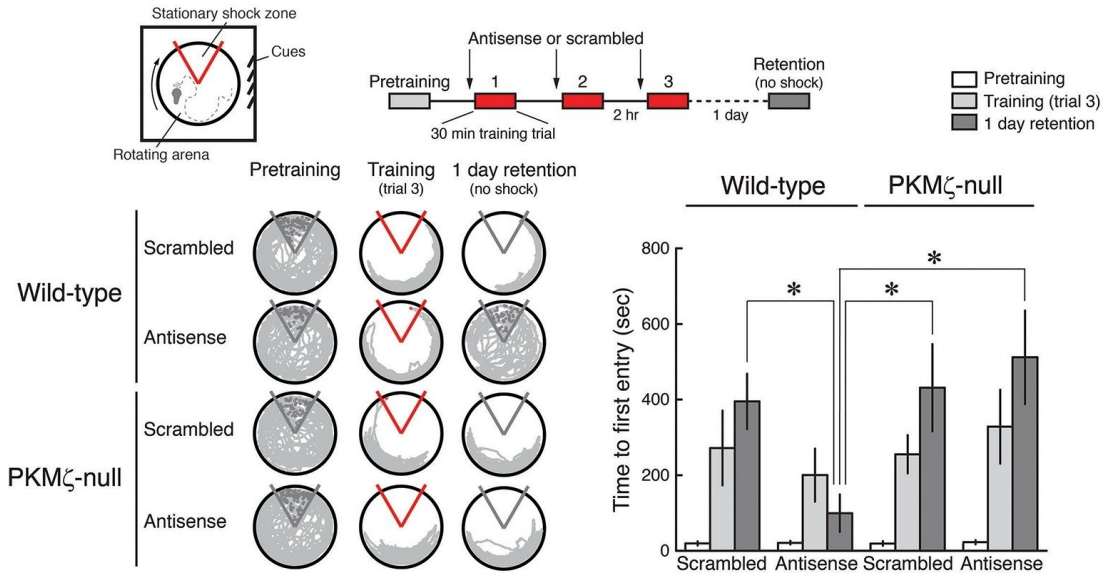
Nebeling et al. (2023), Video 2.

Compensation complicates a single-protein memory claim

PKM-zeta is a persistently active kinase proposed to contribute to late potentiation and memory maintenance. Yet mice lacking it throughout life can retain LTP and perform memory tasks. A study examined whether a related kinase, PKC-iota/lambda, compensated for the missing protein, distinguishing lack of function from substitution by another pathway.

In the tested preparations, a PKM-zeta-directed intervention disrupted relevant plasticity and spatial memory in normal mice but not in knockout mice lacking its target. Conversely, blocking the related kinase disrupted the maintained state more strongly in the knockout condition. The results supported recruitment of a compensatory maintenance route after lifelong loss.

This explains why an absolute statement that one protein is the sole necessary storage switch is too simple. A pathway can contribute in the normal system while another supports a similar outcome after development without it. The mechanism depends on the preparation, timing, and specificity of the intervention rather than the protein’s name alone.



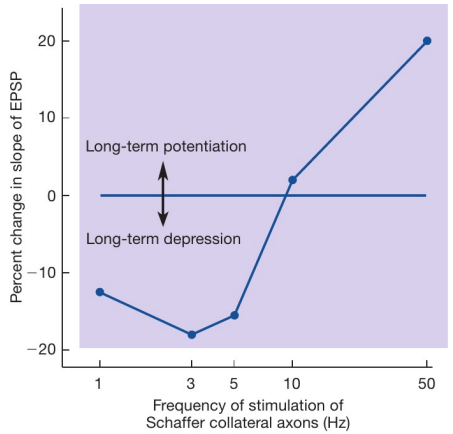
Tsokas et al. (2016), Fig. 4.

Synapses can weaken as well as strengthen

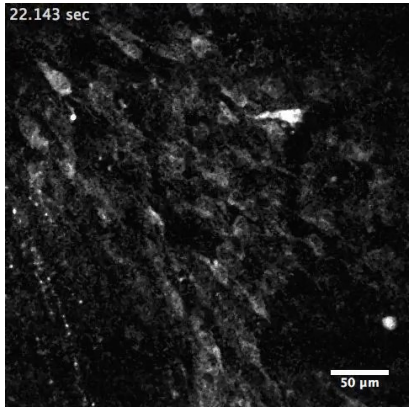
Long-term depression, or LTD, is a persistent reduction in a tested synaptic response after an induction procedure. Different patterns of activity and intracellular signaling can favor depression rather than potentiation. At many excitatory synapses, changes in AMPA-receptor removal or retention contribute to the smaller later response.

Weakening is not simply damage or the absence of learning. Adjusting synaptic strengths in both directions can refine a representation, reduce obsolete influences, and maintain useful response ranges. The cellular route varies across pathways, so one stimulation frequency or receptor mechanism cannot be assumed to produce the same result at every synapse.

A reduced response also does not establish that an entire behavioral memory has been erased. Memories involve distributed connections, and retrieval can fail through several mechanisms. Connecting LTD to a particular learned change requires measurements that identify the affected pathway, the altered behavior, and the timing of the intervention.



Carlson and Birkett (2022), Fig. 13.38, p. 453.



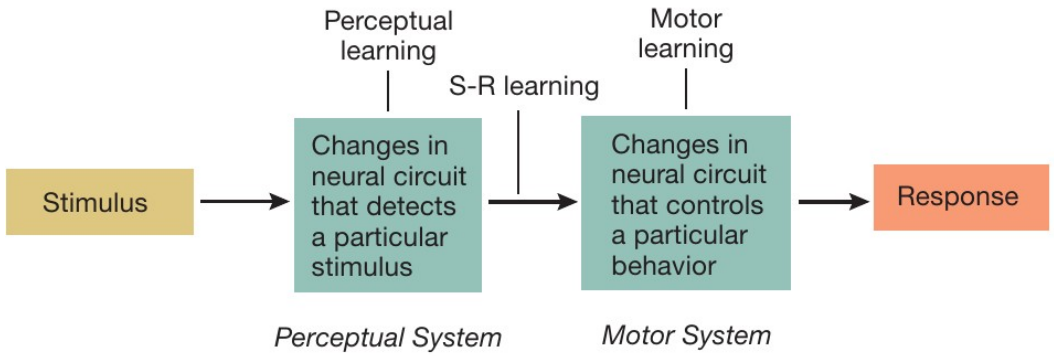
Nebeling et al. (2023), Video 3.

Memory persists through changes across scales

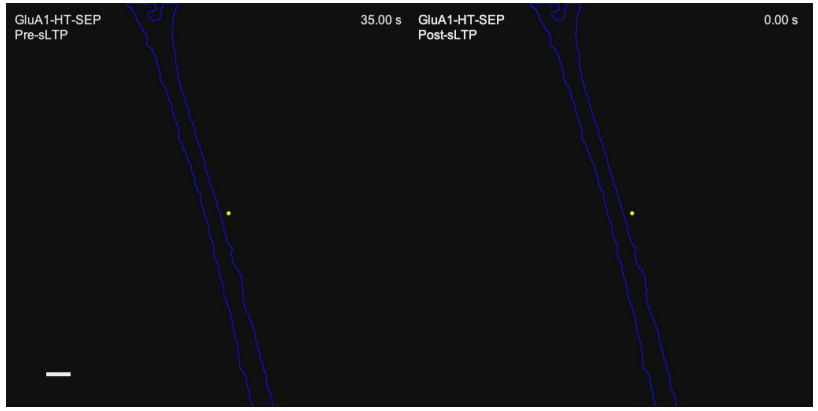
A retained experience can influence perception, action, conscious report, and later learning through different neural systems. Hippocampal relational processing, cortical representations, striatal action learning, and motor circuitry contribute according to what the task demands. Selective preservation after amnesia demonstrates why one behavioral test cannot represent every form of memory.

Within a synapse, coincident activity can trigger calcium entry, enzyme activation, receptor trafficking, and structural modification. Across a network, inhibition and offline reactivation influence which patterns persist and how regions coordinate. These are identifiable biological processes at different scales, and experiments must connect them through their actual measured consequences.

Retrieval adds another requirement: retained information must be accessible and usable under the present conditions. Cues, competing memories, physiological state, and current goals can change its expression. The same stored content can guide different actions, while a temporary failure to express it need not mean that every learning-related cellular change has vanished.



Carlson and Birkett (2022), Fig. 13.6, p. 420.



Wong et al. (2024), fig5-Video 3.

Key takeaways

Learning changes several interacting memory systems. Conditioned responses, skills, perceptual familiarity, and relational memories can change independently; improvement does not always require conscious recollection.

Hippocampal circuits represent relationships and context. Place coding, ensemble activity, and human amnesia show separable contributions to encoding and richly detailed retrieval rather than one universal memory store.

Consolidation changes coordination across networks. Hippocampal ripples, cortical activity, inhibition, and experience influence memory over time; older episodic details can still depend on hippocampal integrity.

Retrieval can preserve or modify what is remembered. Recall depends on cues and ensemble access, and repeated retrieval can alter later competition or updating; failure to express a memory need not prove its erasure.

Synaptic strengthening has identifiable cellular steps. Coincident activity can permit NMDA-dependent calcium entry, activate enzymes, alter AMPA receptor delivery, and change dendritic spine structure.

Lasting memory is more than one molecular switch. Trafficking, actin, inhibitory circuits, protein synthesis, and compensating pathways contribute at different times; a synaptic assay and a remembered event are distinct measurements.